Summary Report, v5: A 100-Metric Library Calibrated on Answers of Known Quality

2026

1 Setup

1.1 Shared setup (identical across versions)

- **Suite.** 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Official test or dev splits only; 150 items per dataset by seeded stratified sampling, 100 flagged for answering (OfficeQA Pro V2: 90; Alaska schema: 93); 3,083 answered items per model. Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % ratio); schema matching is one question per source column with a candidate list; BIRD is graded on the executed result of the gold SQL.
- **Answers.** Eleven models (Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.1 Pro, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b, Llama 3.3 70B); one prompt per item ending in `FINAL ANSWER: <answer>`; temperature 0 where accepted (Claude 5 and GPT-5.5 ran at provider defaults, recorded per row); output budget 700 tokens with mechanical re-asks at 2,500, 8,000, 16,000 and 32,000 for replies cut off before a final line (the budget is counted in each provider's own tokens and, for reasoning models, includes the chain of thought, so it is a termination test rather than an equal text allowance; 55 DeepSeek-R1 answers never terminate at 32,000 and are scored as given); every answer cached with tokens, cost and finish reason. The same cache feeds every version, so leaderboards are directly comparable.
- **Scoring primitives.** Extraction of the final line, Unicode / case / punctuation normalisation, number parsing with sign, percent and scale words, alias lists per gold (best alias wins), list golds, a hedge flag. Native metric per dataset: numeric tolerance (1 %) for numeric golds, exact match for text and boolean, ROUGE-L for free-form.
- **Coverage rule.** A model is in the study only if it covered ≥ 90 % of the suite; a dataset counts only items every model answered and only with ≥ 20 such items.
- **Leaderboard machinery.** Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, SE from the observed information) as the headline pooling; Borda, Kemeny (with cost), Copeland, mean-z, reciprocal-rank fusion and family-balanced z as checks; 1,000 item bootstraps of the Borda and Bradley–Terry ranks; Friedman with Iman–Davenport, Nemenyi CD, Holm-corrected pairwise Wilcoxon across datasets; paired permutation and Wilcoxon per item within datasets (Holm); Kendall's W among datasets; majority-class baselines for boolean sets; post-stratified official-split estimates with a weighted bootstrap and positive-class F1 at the official match ratio; floor / ceiling / self-stability diagnostics per dataset.
- **Sampled designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples on the weight simplex at step 0.05 (20 seeds each) with centred L_2 discrepancy and nearest-neighbour

distances, compared with the exhaustive optimum where one exists.
- **Reporting.** Each version's `run_all.sh` regenerates its `REPORT.md` from the cached answers; every number in this document is read from those output files.

1.2 What v5 searches

- **Metric library.** 100 per-answer metrics in twelve families (exact-match variants, token overlap, character and sequence similarity, numeric tolerance and graded numeric error, set metrics for list golds, structure and task-specific scores, length, commitment / hedging, semantic placeholders), computed for every (model, item) pair and for every anchor.
- **Anchors as the quality signal.** 31,380 synthetic answers generated from the golds by 20 typed operators with a known utility each (identity, verbose, tagged, unit variant = 1; paraphrase 0.95; near misses 0.75 / 0.45 / 0.15; superset 0.6; typo 0.6; partial 0.5; shuffled 0.4; hedge 0.3; list dump 0.25; digit typo 0.2; sign flip and magnitude 0.05; flip, no answer, wrong random = 0). v5 is the only version whose objective is correctness against this signal rather than agreement among the ranked models.
- **Selection.** Drop constant metrics; cluster |Spearman| ≥ 0.95 into representatives; keep candidates above an anchor-agreement floor; forward selection maximising leave-one-dataset-out Spearman of a non-negative least-squares fit to utility.
- **Weights.** Exhaustive grid at step 0.02 over the selected weights (sum to 1) $\times$ five hedge penalties λ (score multiplied by $1 - \lambda$ when the answer hedges): 6,630 points. Objective $J = 0.30\,\rho + 0.15\,\mathrm{AUC} + 0.20\,(1 - \mathrm{JS}/\ln 2) + 0.15\,\tau_{\mathrm{native}} + 0.20\,(1 - \text{wrong mean})$, where JS is the Jensen–Shannon divergence between datasets' score distributions at matched operator (cross-dataset consistency) and τ_{native} the agreement of the composite's model ranking with the native-metric ranking. The ensemble is the mean of the 100 best weightings, so each answer carries a score and a spread.
- **Validation.** Leave-one-dataset-out re-selection and re-fitting; utility-scale perturbation (± 0.15, five draws); gaming probes (mean score per operator vs utility against single-metric baselines); sampled designs looked up in the exhaustive grid.
- **Leaderboards.** Exact match, the native metric, the ensemble (headline) and the five best single weightings, each pooled by Bradley–Terry with the full significance battery.

2 Results

2.1 Three metrics carry the library's information

Of 100 metrics, 45 are distinct at |Spearman| ≥ 0.95 and forward selection stops at three: token F1, numeric decay and prefix ratio (Figures 1, 2). Their NNLS fit reaches leave-one-dataset-out ρ 0.875 against 0.814 for the native typed metric and 0.831 for exact match.

2.2 The calibrated ensemble

The grid is decay-dominated (Figures 4, 5): the top 100 put 0.75 on numeric decay with a hedge penalty of 0.63. The ensemble reaches ρ 0.904 and AUC 0.996 on the anchors, above every single-metric baseline (Table 8, Figure 6); the probes show where the gain comes from: near misses, unit variants and partial answers get graded credit instead of zero (Figure 7, Table 36). The sampled designs land within 0.0015 of the exhaustive best (Table 9).

Table 1: v5 audit summary.

Library	100 metrics, 10 constant on this data, 45 cluster representatives, 28 candidates above the floor
Selected	tok_f1, num_decay_2p5, prefix_ratio; LODO ρ of the selected set 0.875 vs native typed metric alone 0.814 and exact match alone 0.831
Grid	6,630 weightings (step 0.02 × 5 λ); J top-1 0.934, top-100 mean 0.932, grid median 0.924; range 0.877–0.934
Ensemble weights	tok_f1 0.17, num_decay_2p5 0.75, prefix_ratio 0.07, $\lambda = 0.63$; decay weight spans 0.52–0.96 across the top 100
Ensemble terms	ρ 0.904, AUC 0.996, JS 0.054, τ native 0.893, wrong mean 0.032
Baselines	token F1 ρ 0.859 / J 0.906; numeric tolerance ρ 0.806 / J 0.909; ensemble ρ 0.904 / J 0.933
Sampled designs	all three within 0.0015 of the exhaustive best (0.9337)
Leave-one-dataset-out	mean held-out ρ 0.894 (re-fitted) vs 0.895 (full); same metric set selected in 90 % of folds
Utility perturbation	mean L_1 shift of the ensemble weights 0.042; selection changed in 0 % of draws
Probes (MAE to utility)	ensemble 0.113, token F1 0.144, native 0.155, exact match 0.199
Leaderboard (ensemble, BT)	first Gemini 3.8 Flash (P(first) 0.74), then Claude Opus 5, GPT-5.5; last Llama 3.3 70B; Friedman p = 2.0e-13; $W = 0.26$
Other rules	native metric: Gemini 3.8 Flash first; exact match: Claude Opus 5 first; τ ensemble vs native 0.78, vs exact match 0.85

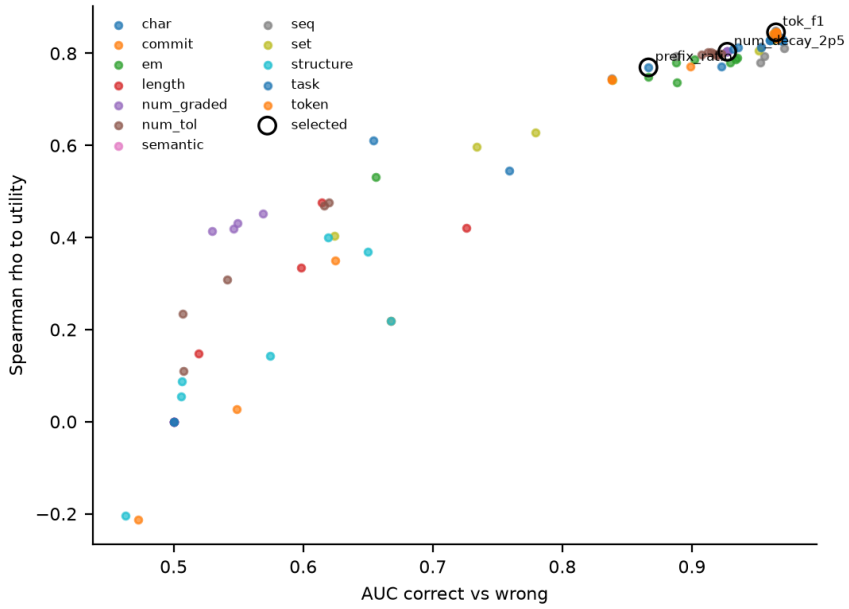


Figure 1: The 100-metric library on the anchors: AUC against Spearman ρ, coloured by family; the three selected metrics are circled.

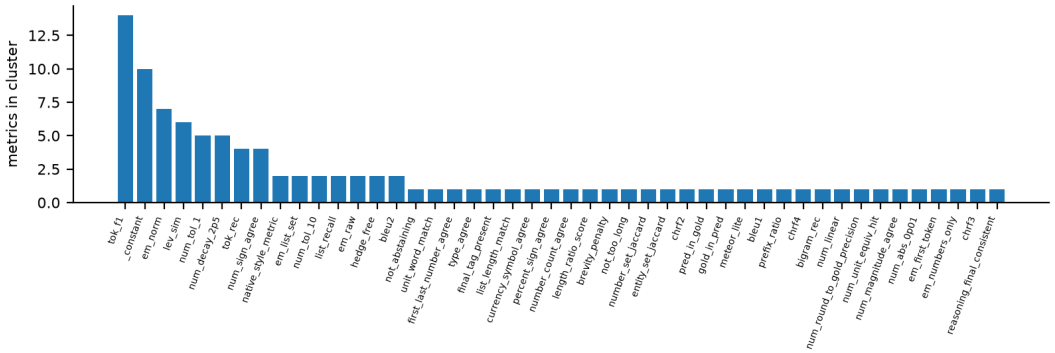


Figure 2: Cluster sizes after merging metrics with $|\text{Spearman}| \geq 0.95$ (45 representatives).

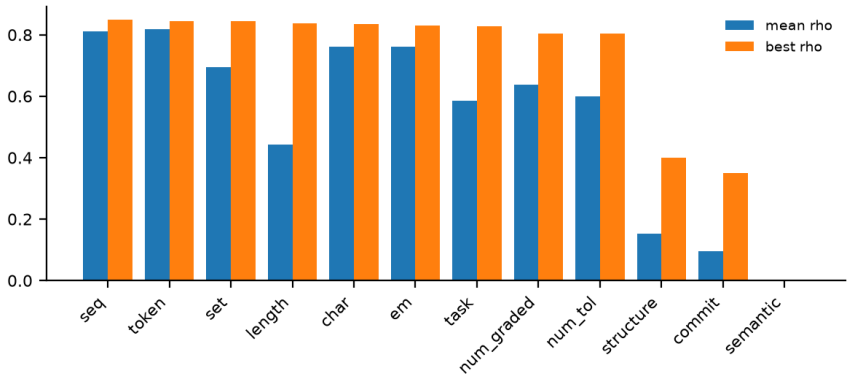


Figure 3: Anchor ρ per metric family: mean and best member.

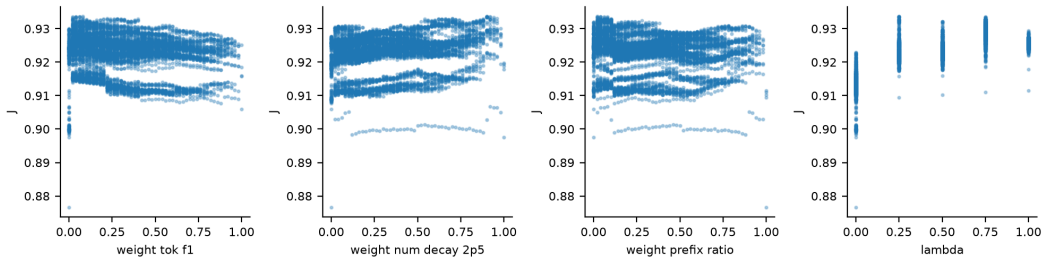


Figure 4: Objective J against each selected weight and the hedge penalty λ over all 6,630 grid points.

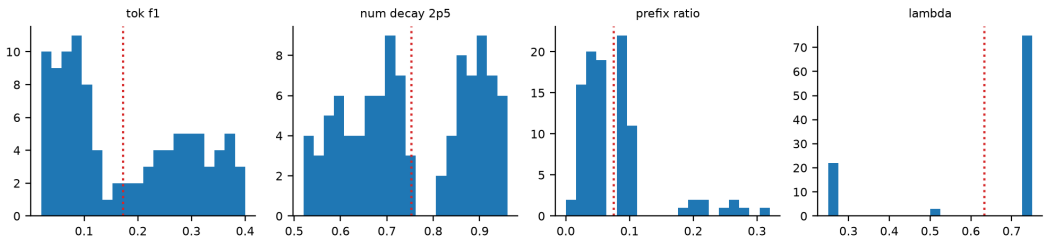


Figure 5: Weights and λ of the 100 best grid points that form the ensemble (dotted = mean).

Table 2: Baselines under the same objective.

baseline	rho	auc	js	bad	tau_native	J
native_style_metric	0.826	0.935	0.031	0.014	0.992	0.925
em_norm	0.849	0.963	0.024	0.012	0.916	0.927
tok_f1	0.859	0.965	0.079	0.032	0.884	0.906
rouge_l	0.862	0.965	0.082	0.032	0.884	0.906
num_tol_1	0.806	0.919	0.032	0.016	0.942	0.909
uniform_over_selected_no_gate	0.882	0.970	0.098	0.028	0.907	0.912
ensemble_top100	0.904	0.996	0.054	0.032	0.893	0.933

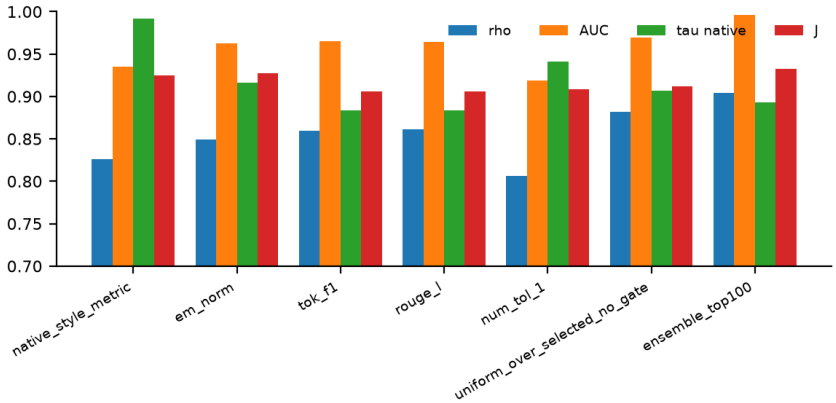


Figure 6: Single-metric baselines against the ensemble under the same objective terms.

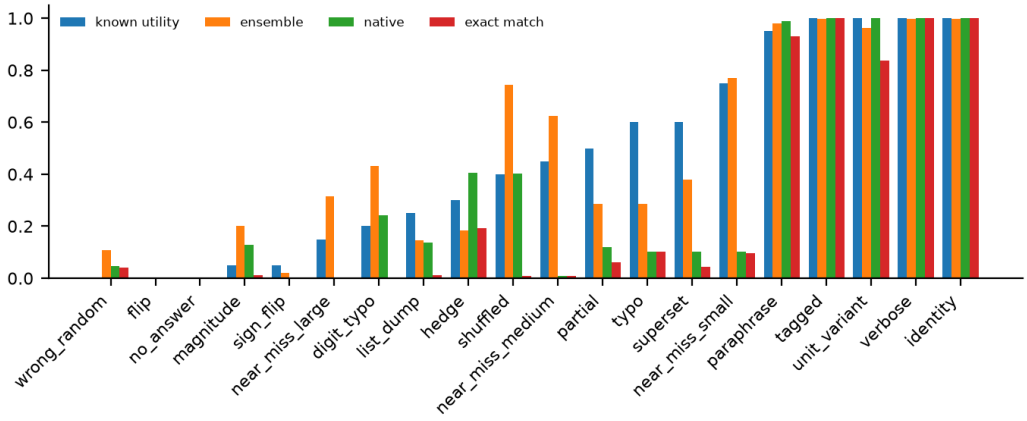


Figure 7: Mean score per anchor operator under the ensemble, the native metric and exact match, against known utility.

Table 3: Sampled designs vs the exhaustive grid (best $J = 0.9337$ over 6630 points).

design	best_J_mean	best_J_sd	regret_mean	regret_max	min_pairwise_L1	discrepancy	zero_share
dirichlet_lattice	0.9333	0.0004	0.0004	0.0010	0.0400	0.2728	0.0615
lhs_simplex	0.9331	0.0003	0.0006	0.0011	0.0400	0.2702	0.0555
uniform_lattice	0.9332	0.0004	0.0005	0.0015	0.0400	0.2639	0.1115

2.3 Validation

Leaving any dataset out re-selects the same three metrics in 90 % of folds and changes held-out ρ by 0.001 on average (Figure 8); perturbing the utility scale by ± 0.15 moves the weights by 0.042 in L_1 and never changes the selection. The hand-set utility scale is the main assumption, and these checks say the result is not sensitive to it.

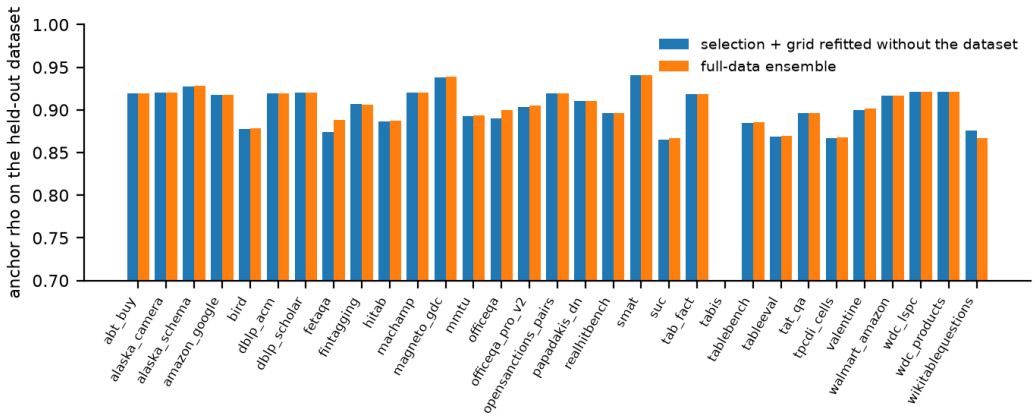


Figure 8: Leave-one-dataset-out: anchor ρ on the held-out dataset under the re-fitted and the full-data ensemble.

Table 4: Validation summary.

quantity	value
lodo_mean_rho_lodo	0.8940
lodo_mean_rho_full	0.8949
lodo_same_selection_share	0.9032
severity_l1_shift_mean	0.0419
severity_l1_shift_max	0.0432
severity_selection_changed_share	0.0000
probe_mae_ensemble	0.1132
probe_mae_em	0.1991
probe_mae_native	0.1545
probe_mae_tok_f1	0.1440

2.4 Leaderboard

Under the ensemble pooled by Bradley–Terry, Gemini 3.8 Flash is first, Claude Opus 5 and GPT-5.5 next, Llama 3.3 70B last; Borda puts Claude Opus 5 first. The five best single weightings give the same ranking as the ensemble ($\tau = 1.00$); the native metric and exact match move the top three. Dataset agreement stays moderate ($W = 0.26$). The ensemble spread per answer (Figure 11) is largest on the free-form and list datasets, which is where the weighting matters.

Table 5: Leaderboard under the headline rule (ensemble).

rank	model	BT strength	SE	Borda rank	Borda 95%	P(first) BT	mean score
1	Gemini 3.8 Flash	0.118	0.015	3	[1, 4]	0.740	0.834
2	Claude Opus 5	0.111	0.000	1	[1, 3]	0.260	0.834
3	GPT-5.5	0.034	0.015	2	[1, 3]	0.000	0.823
4	Claude Sonnet 5	0.029	0.015	4	[4, 7]	0.000	0.815
5	DeepSeek-R1	0.029	0.015	5	[3, 7]	0.000	0.815
6	Gemini 3.1 Pro	0.022	0.015	6	[4, 8]	0.000	0.811
7	Grok 4	0.012	0.015	7	[4, 8]	0.000	0.810
8	GPT-5.4-mini	-0.024	0.015	8	[6, 9]	0.000	0.804
9	DeepSeek-chat	-0.035	0.015	9	[7, 9]	0.000	0.797
10	gpt-oss-120b	-0.103	0.015	10	[10, 11]	0.000	0.783
11	Llama 3.3 70B	-0.193	0.015	11	[10, 11]	0.000	0.747

Table 6: Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	DeepSeek-R1	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
exact match	2 (3)	1 (1)	3 (2)	7 (5)	4 (4)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	1.5e-15	0.28
native metric	1 (3)	3 (2)	2 (1)	7 (5)	5 (4)	4 (6)	6 (7)	9 (8)	8 (9)	10 (10)	11 (11)	4.9e-15	0.27
ensemble	1 (3)	2 (1)	3 (2)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	2.0e-13	0.26
top1	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	4.2e-13	0.25
top2	1 (3)	2 (1)	3 (2)	5 (5)	4 (4)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	5.5e-13	0.25
top3	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	4.2e-13	0.25
top4	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	4.2e-13	0.25
top5	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	3.7e-13	0.25

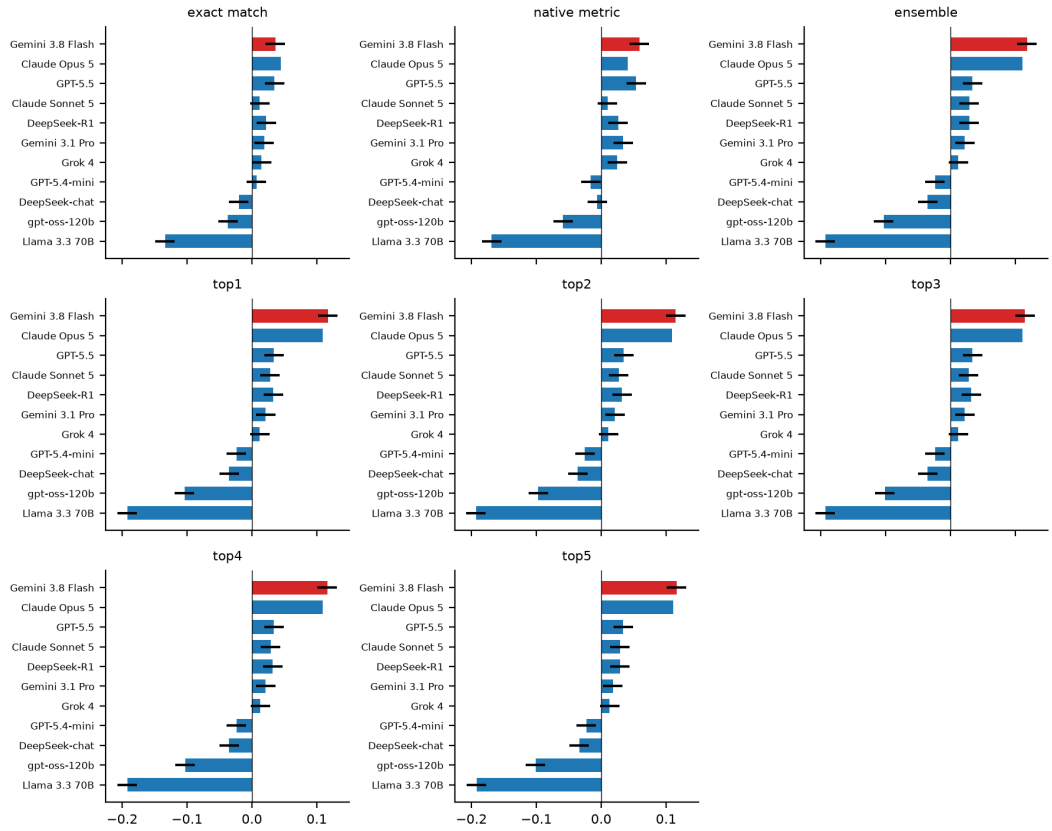


Figure 9: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

2.5 Comparability and diagnostics

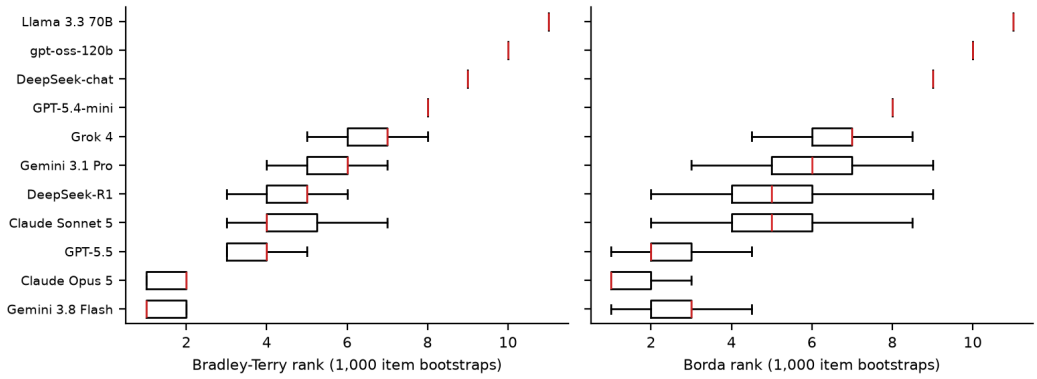


Figure 10: Bootstrap distribution of each model's rank under the headline rule (ensemble).

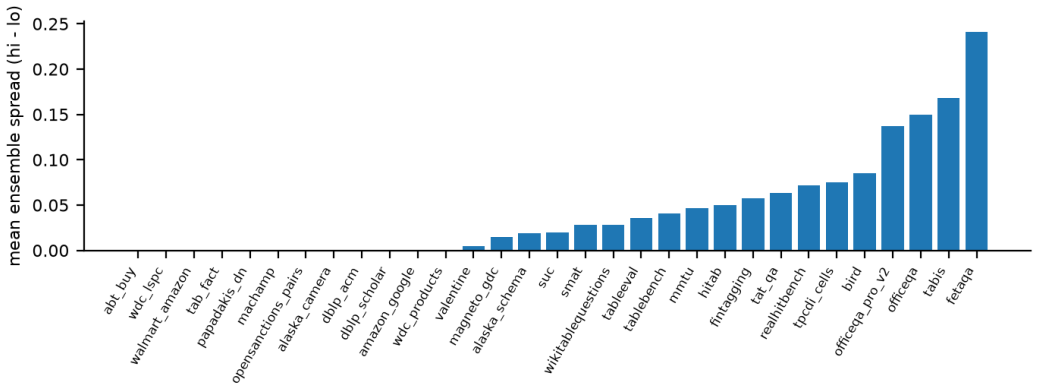


Figure 11: Ensemble uncertainty per dataset: mean spread of the per-answer score across the 100 weightings.

Table 7: Official-split estimates under the headline rule averaged over the ten models.

dataset	suite_mean	official_ estimate	suite_f1_ pos	official_precision_ pos	official_recall_ pos	official_f1_ pos	official_positive_ share
abt_buy	0.971	0.987	0.969	0.933	0.950	0.940	0.108
alaska_camera	0.945	0.945	0.939	1.000	0.895	0.939	0.500
alaska_schema	0.911	0.911	–	–	–	–	–
amazon_google	0.935	0.932	0.927	0.609	0.940	0.737	0.102
bird	0.255	0.274	–	–	–	–	–
dblp_acm	0.987	0.992	0.986	0.981	0.980	0.978	0.180
dblp_scholar	0.948	0.966	0.947	0.901	0.922	0.909	0.186
fetaqa	0.555	0.555	–	–	–	–	–
fintagging	0.830	0.830	–	–	–	–	–
hitab	0.776	0.790	–	–	–	–	–
machamp	0.887	0.891	0.892	0.649	0.852	0.735	0.175
magneto_gdc	0.919	0.922	–	–	–	–	–
mmtu	0.867	0.816	–	–	–	–	–
officeqa	0.286	0.286	–	–	–	–	–
officeqa_pro_v2	0.312	0.312	–	–	–	–	–
opensanctions_ pairs	0.835	0.748	0.790	0.993	0.677	0.795	0.769
papadakis_dn	0.940	0.987	0.831	0.527	0.967	0.662	0.018
realhitbench	0.721	0.720	–	–	–	–	–
smat	0.889	0.887	–	–	–	–	–
suc	0.952	0.951	–	–	–	–	–
tab_fact	0.802	0.792	0.761	0.903	0.674	0.767	0.500
tabis	0.879	0.884	–	–	–	–	–
tablebench	0.860	0.865	–	–	–	–	–
tableeval	0.775	0.778	–	–	–	–	–
tat_qa	0.798	0.756	–	–	–	–	–
tpcdi_cells	0.417	0.449	–	–	–	–	–
valentine	0.983	0.977	–	–	–	–	–
walmart_amazon	0.959	0.974	0.949	0.824	0.931	0.872	0.094
wdc_lspc	0.982	0.984	0.983	0.968	0.978	0.972	0.264
wdc_products	0.978	0.985	0.979	0.982	0.967	0.974	0.280
wikitablequestions	0.851	0.850	–	–	–	–	–

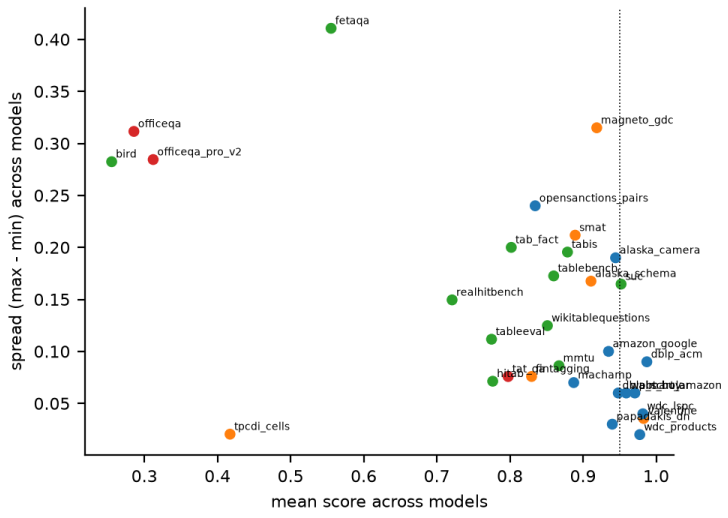


Figure 12: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

A All figures

Every figure that can be drawn from the saved outputs; source data in `data/`.

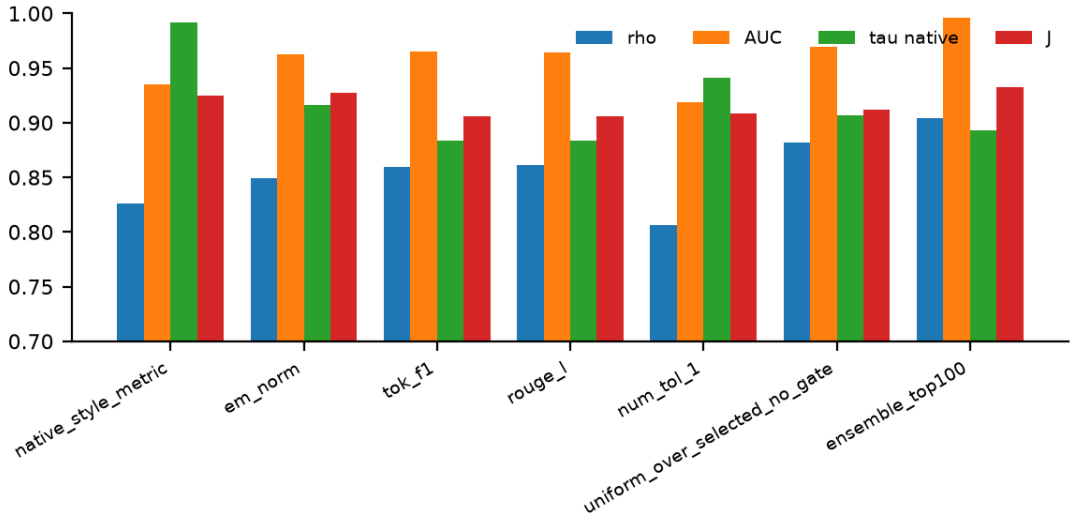


Figure 13: Single-metric baselines against the ensemble under the same objective terms.

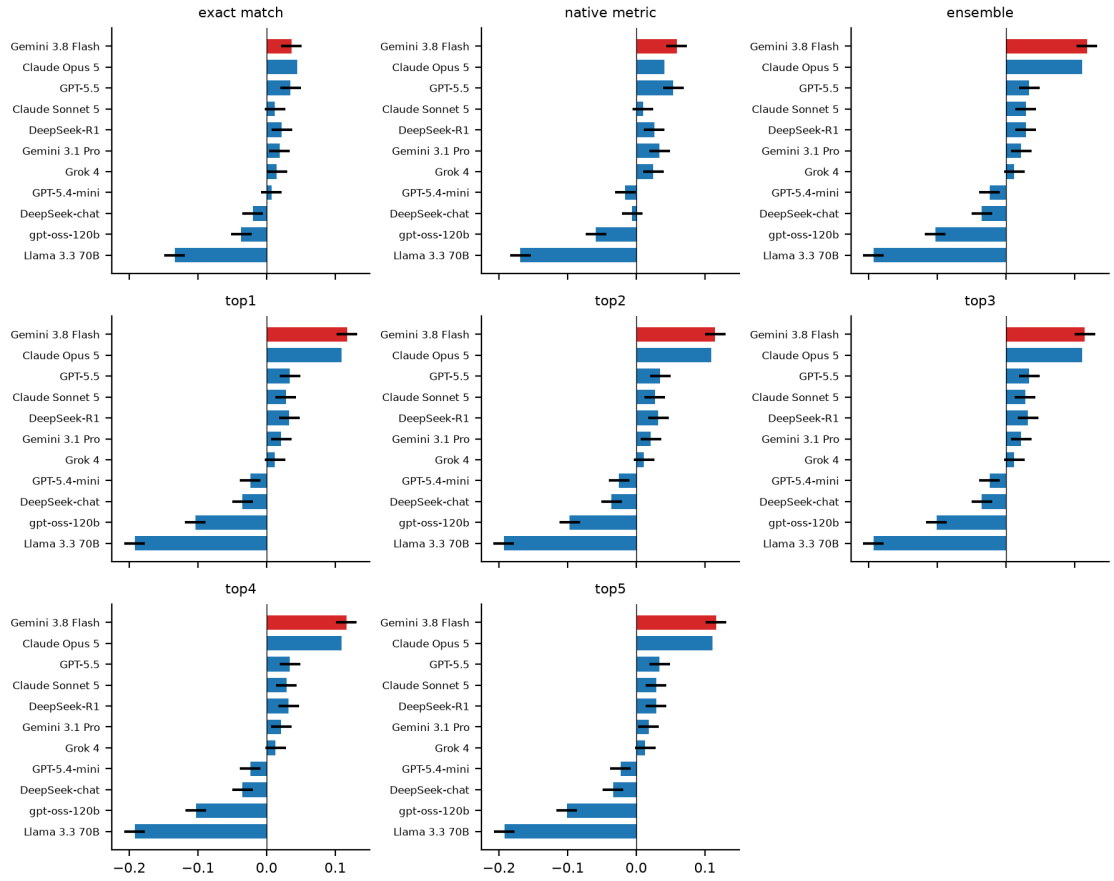


Figure 14: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

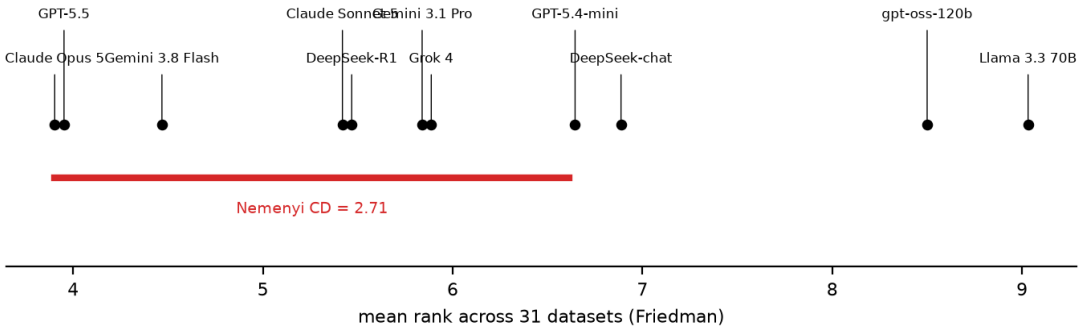


Figure 15: Critical-difference view of the Friedman test under the headline rule: mean rank per model and the Nemenyi critical difference at $\alpha = .05$.

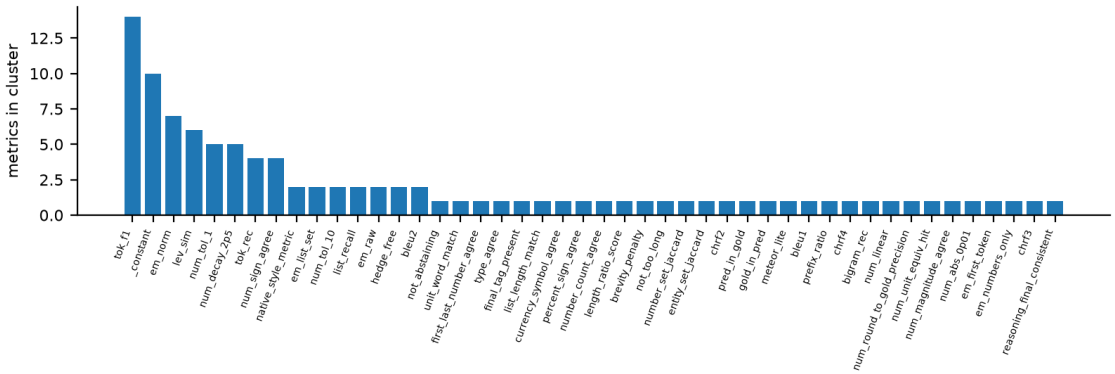


Figure 16: Cluster sizes after merging metrics with |Spearman| ≥ 0.95 (45 representatives).

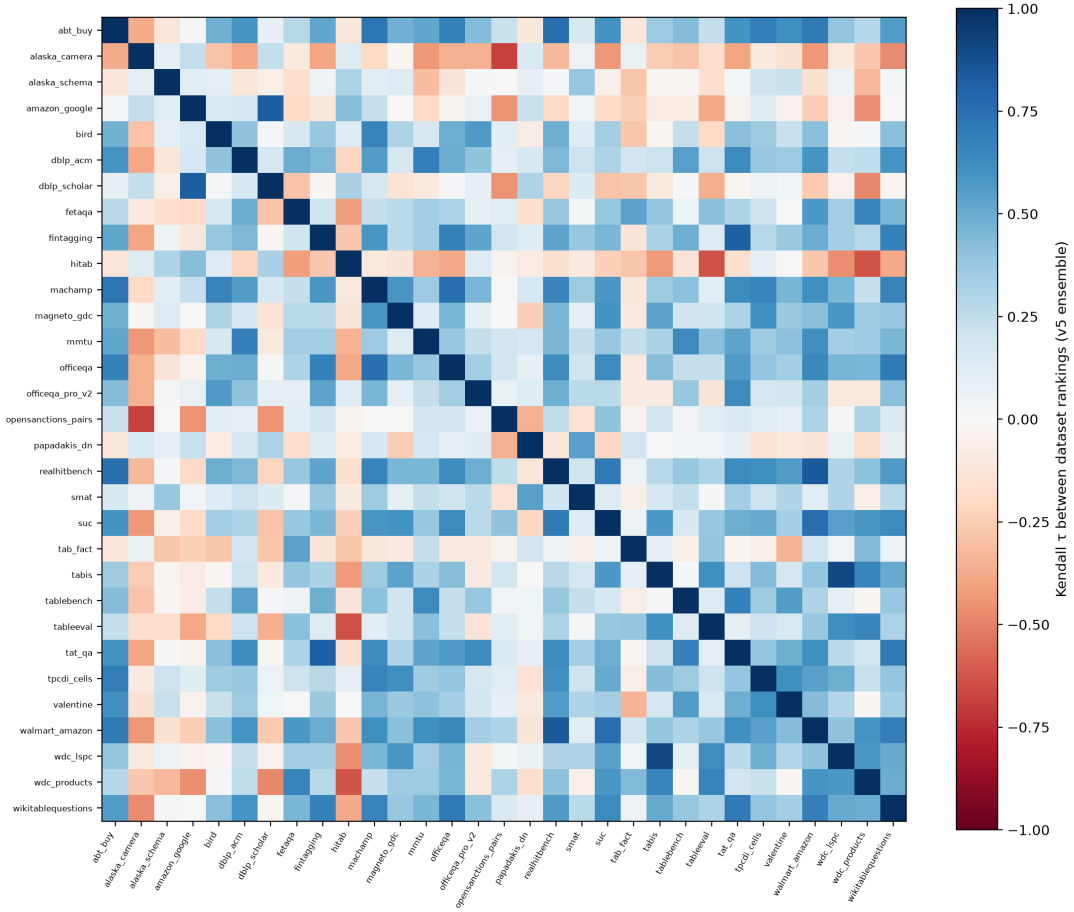


Figure 17: Kendall τ between the model rankings of each pair of datasets under the headline rule.

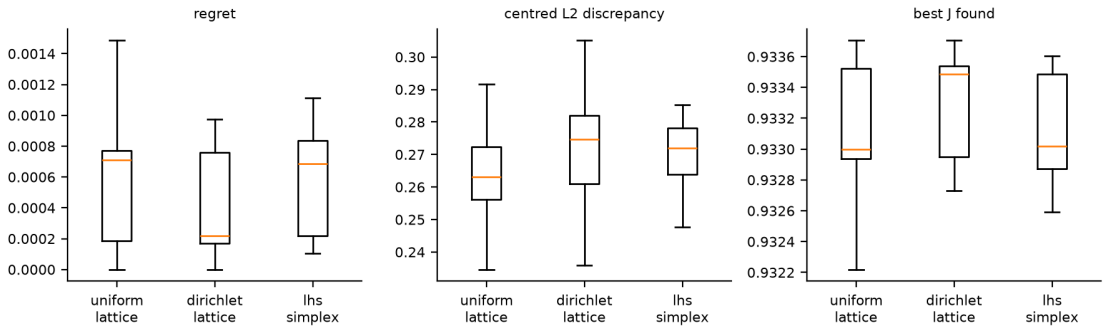


Figure 18: Sampled designs over 20 seeds against the exhaustive grid.

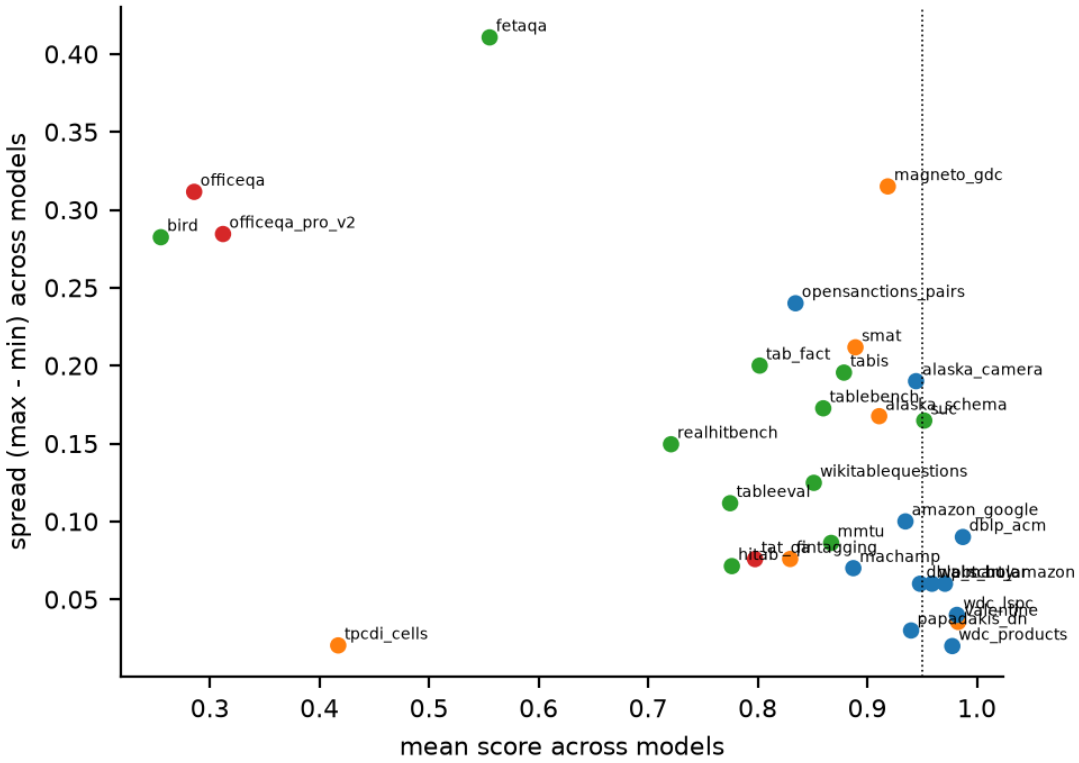


Figure 19: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

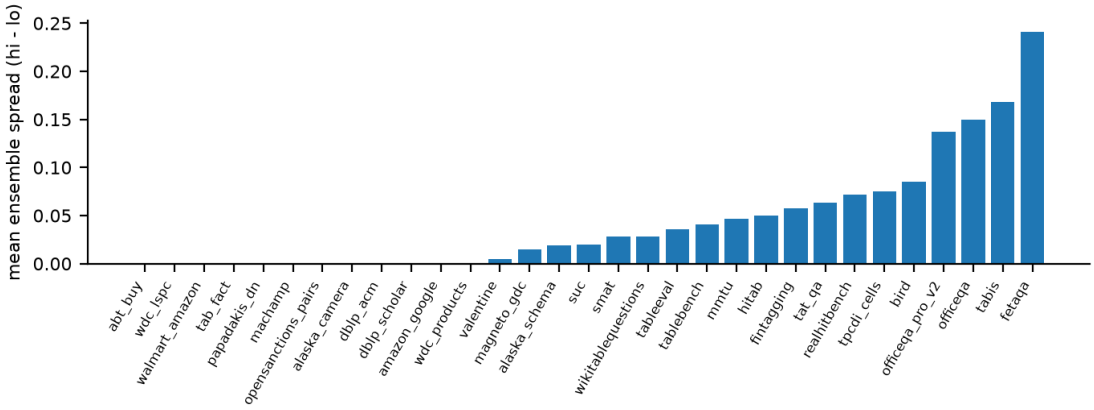


Figure 20: Ensemble uncertainty per dataset: mean spread of the per-answer score across the 100 weightings.

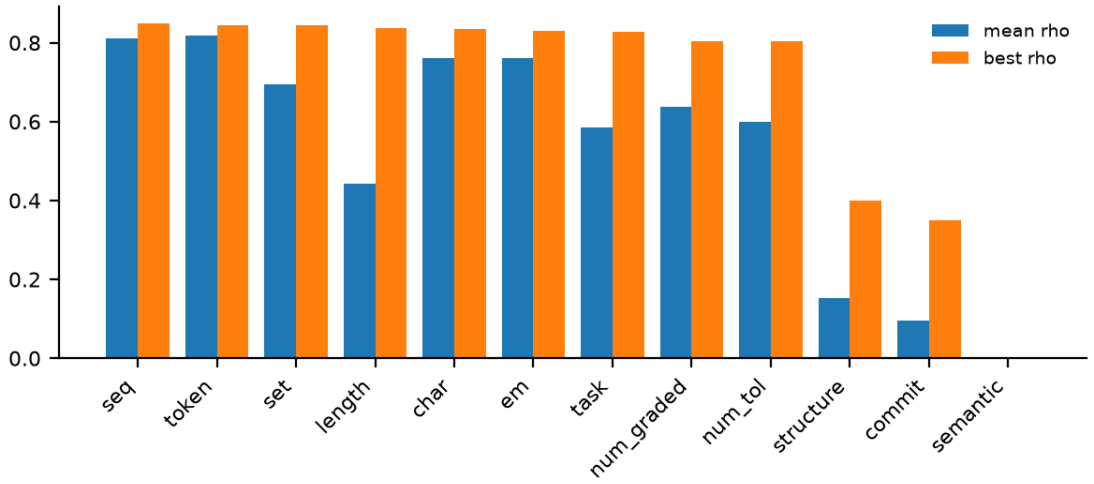


Figure 21: Anchor ρ per metric family: mean and best member.

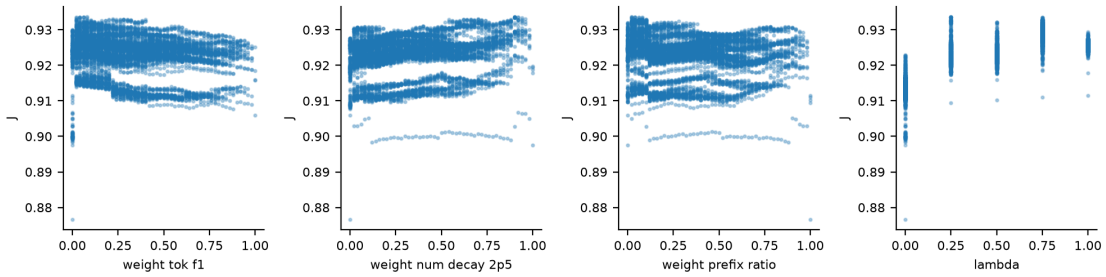


Figure 22: Objective J against each selected weight and the hedge penalty λ over all 6,630 grid points.

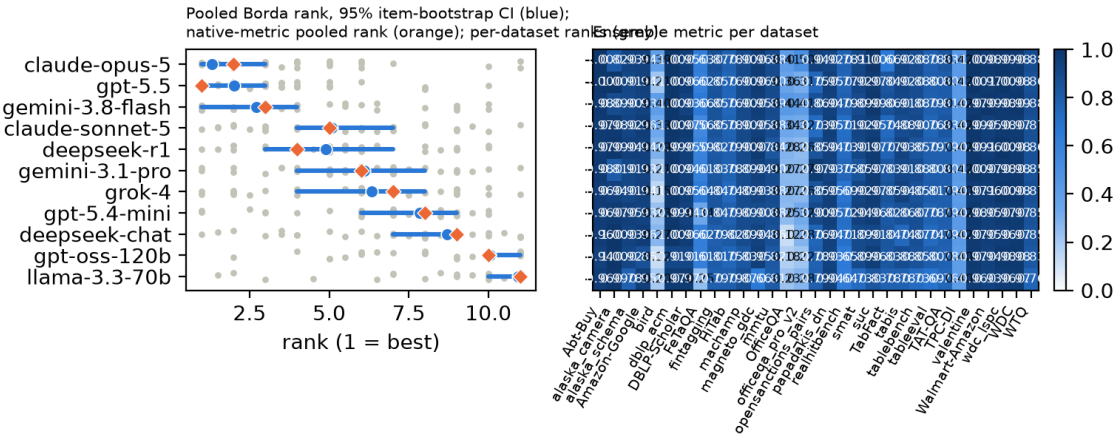


Figure 23: Ensemble leaderboard with bootstrap intervals and the native-metric ranking beside.

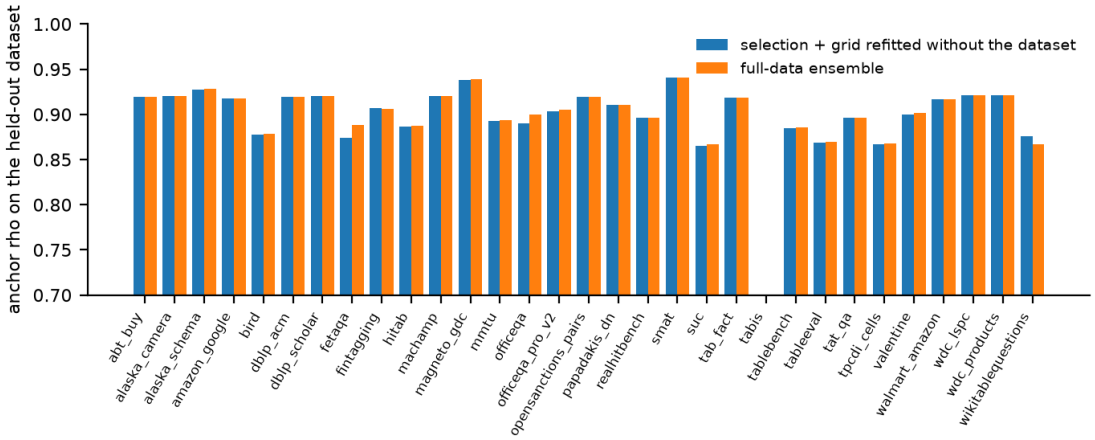


Figure 24: Leave-one-dataset-out: anchor ρ on the held-out dataset under the re-fitted and the full-data ensemble.

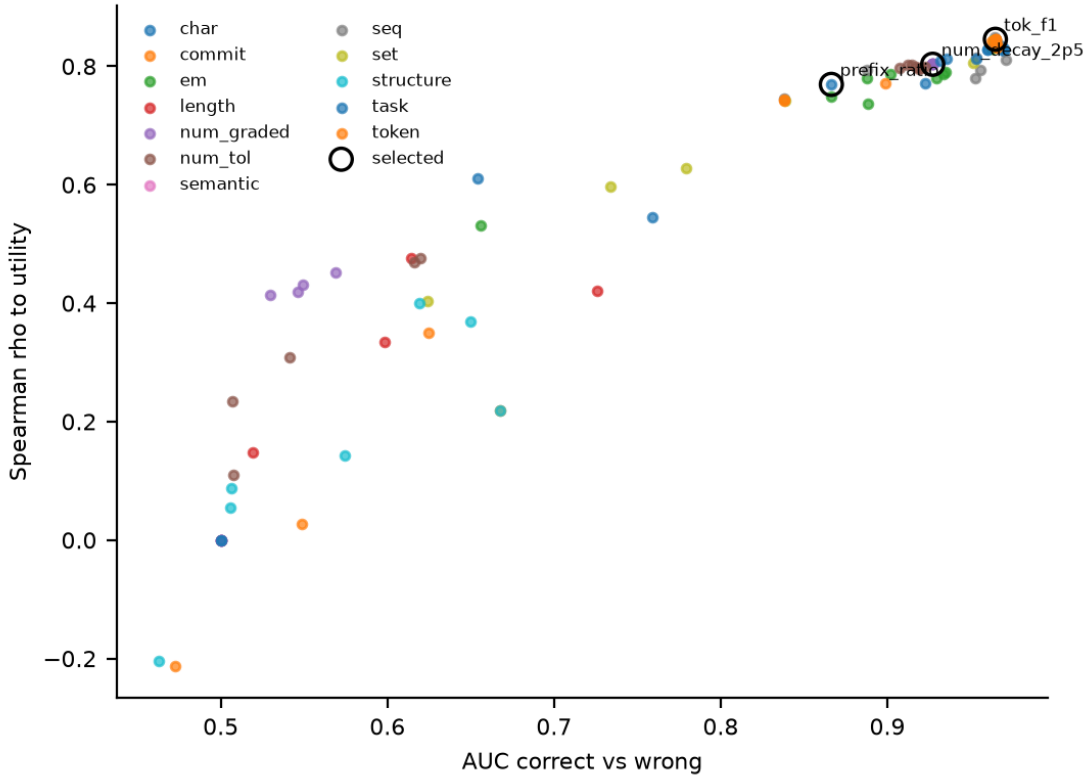


Figure 25: The 100-metric library on the anchors: AUC against Spearman ρ, coloured by family; the three selected metrics are circled.

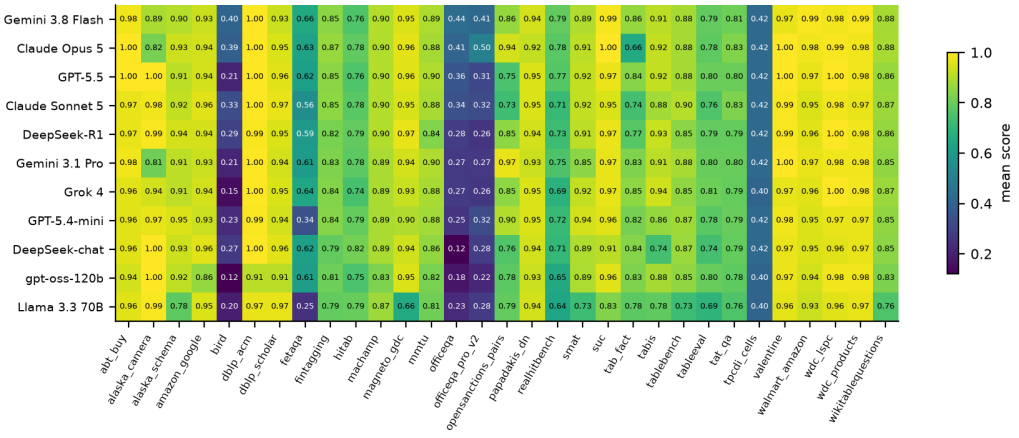


Figure 26: Mean score per model and dataset under the headline rule.

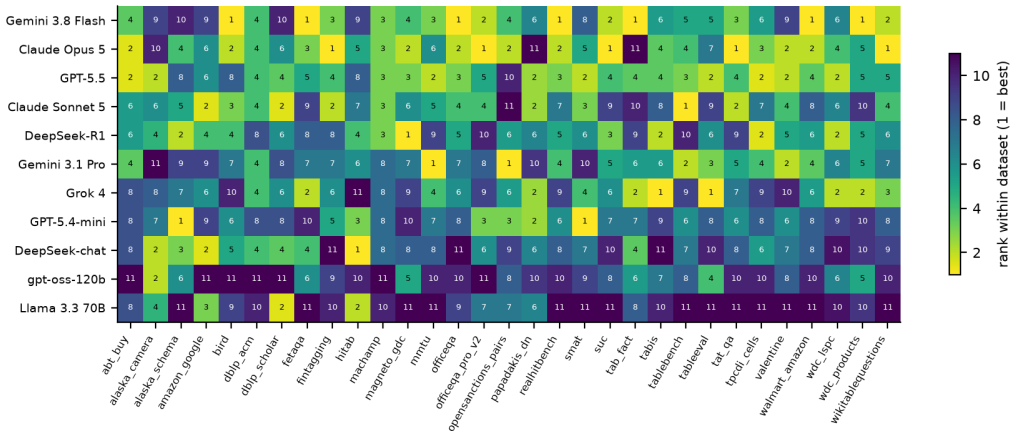


Figure 27: Rank per model and dataset under the headline rule.

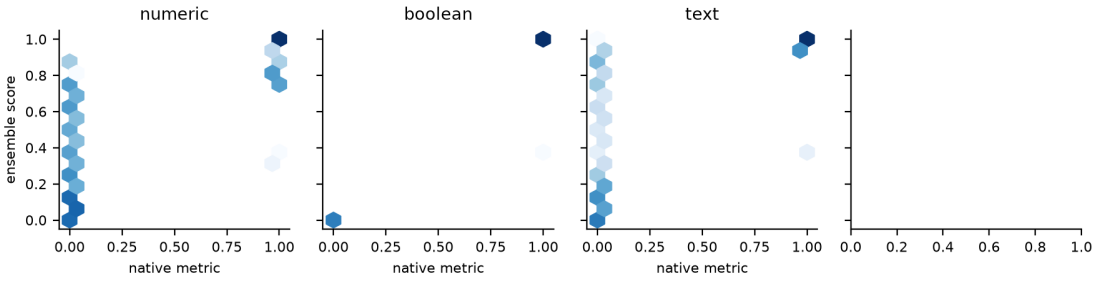


Figure 28: Per-answer native metric against the ensemble score, by answer type.

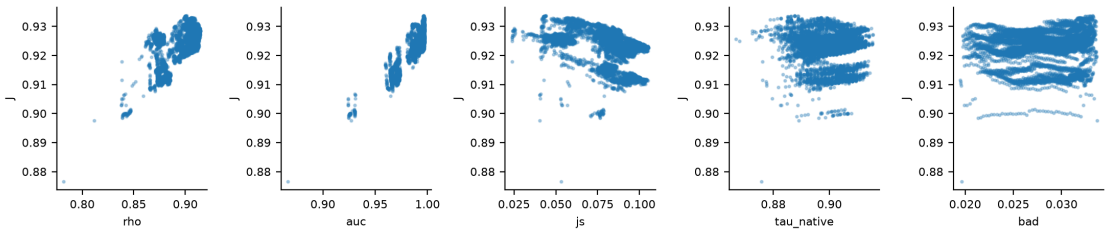


Figure 29: Objective J against each of its five terms over the grid.

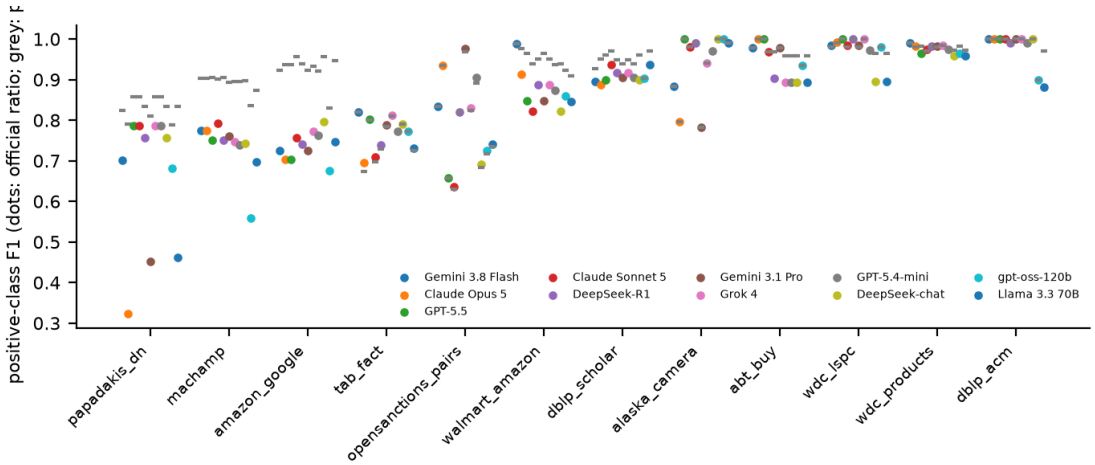


Figure 30: Positive-class F1 on the balanced pool (grey) and at the official match ratio (coloured) per model and matching dataset.

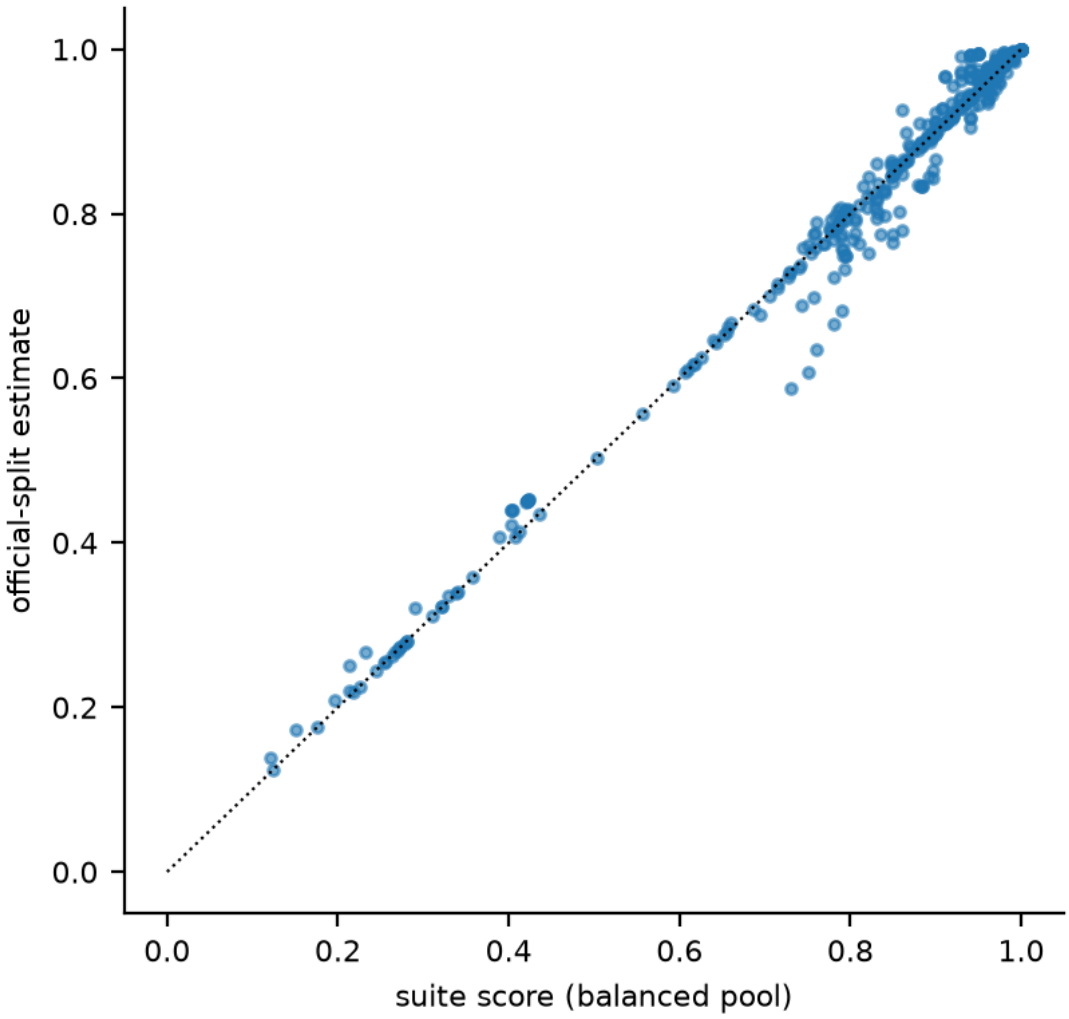


Figure 31: Suite score against post-stratified official-split estimate for every (dataset, model) pair under the headline rule.

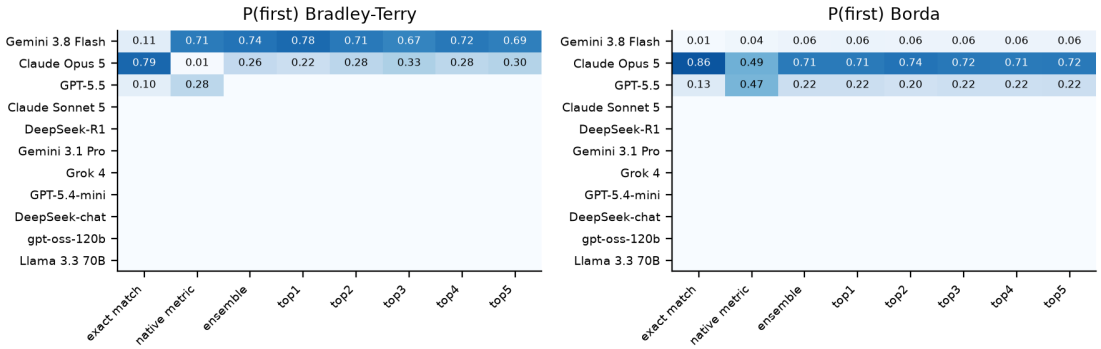


Figure 32: Bootstrap probability of first place under every rule, by pooling method.

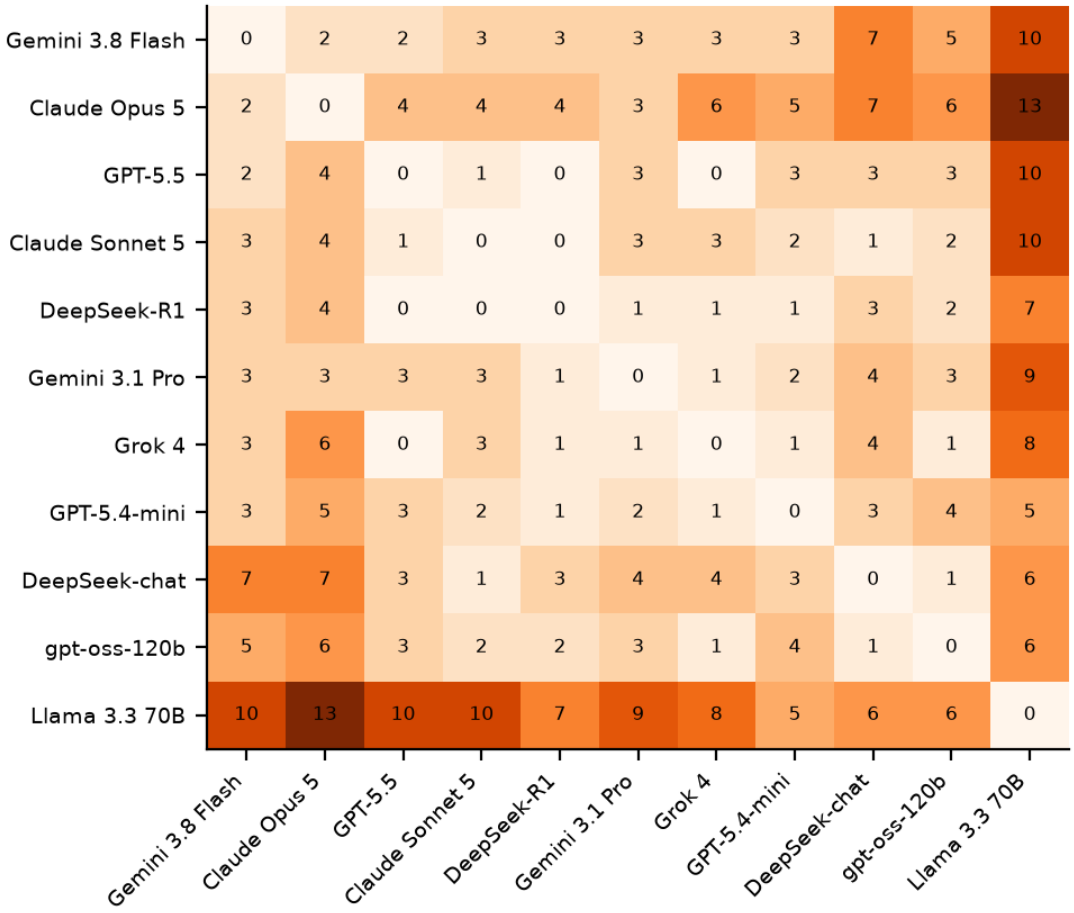


Figure 33: Number of datasets (of 31) in which the per-item paired test separates each pair of models under the headline rule (Holm within dataset).

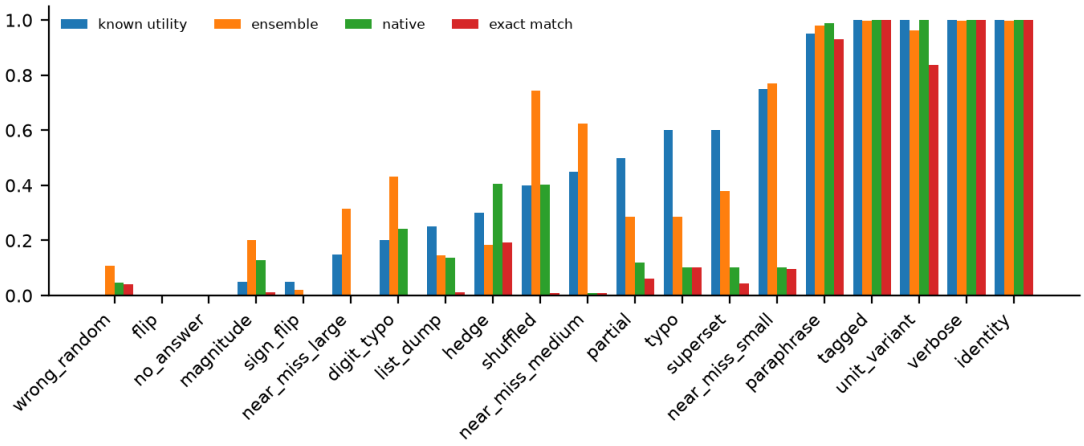


Figure 34: Mean score per anchor operator under the ensemble, the native metric and exact match, against known utility.

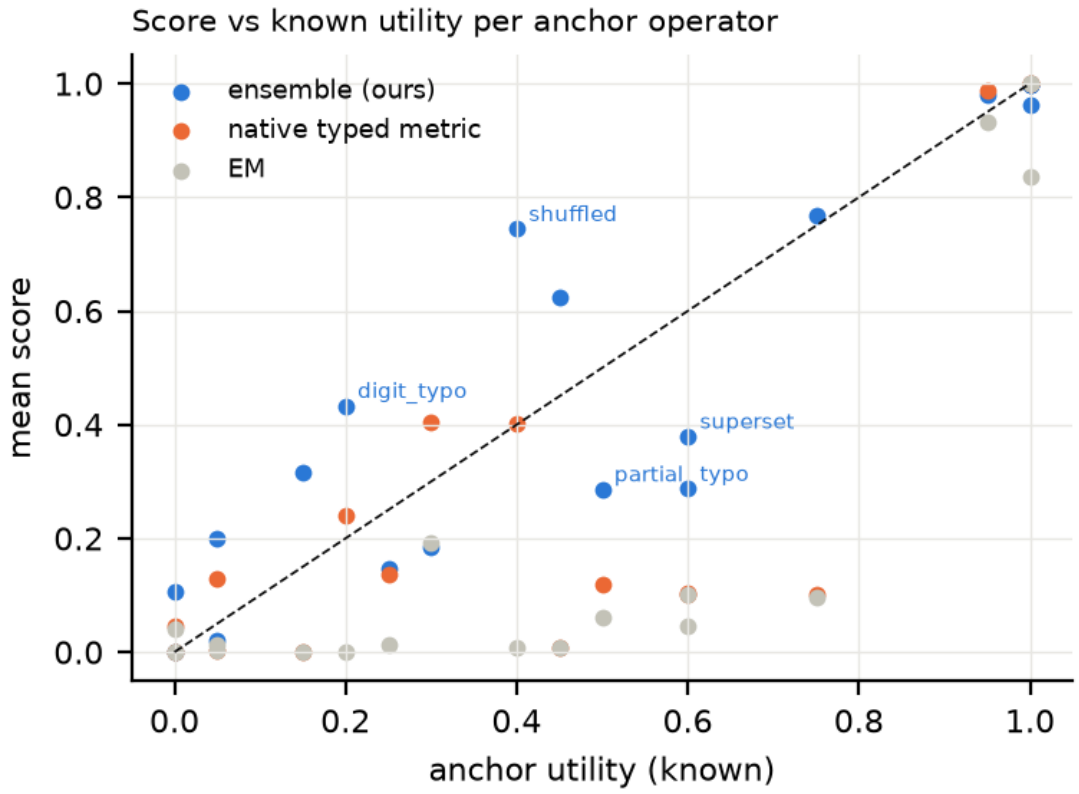


Figure 35: Probes: mean score per anchor operator against known utility for the ensemble and the single-metric baselines.

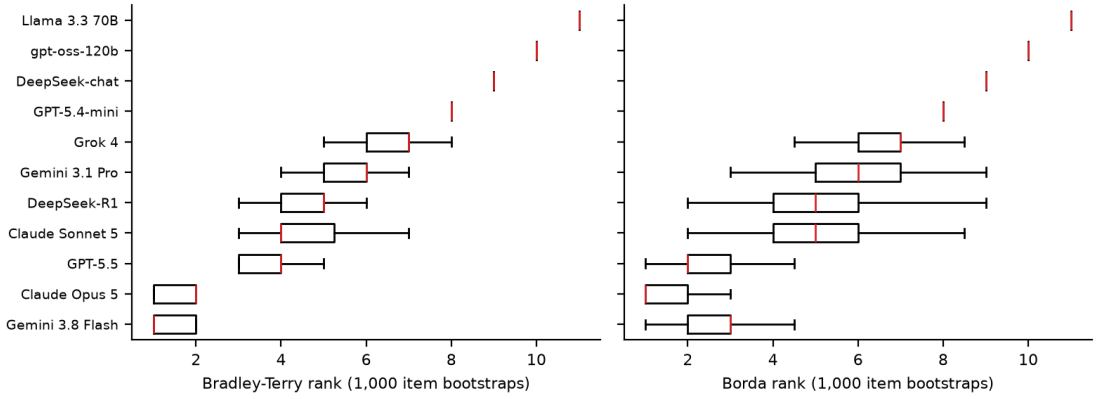


Figure 36: Bootstrap distribution of each model's rank under the headline rule (ensemble).

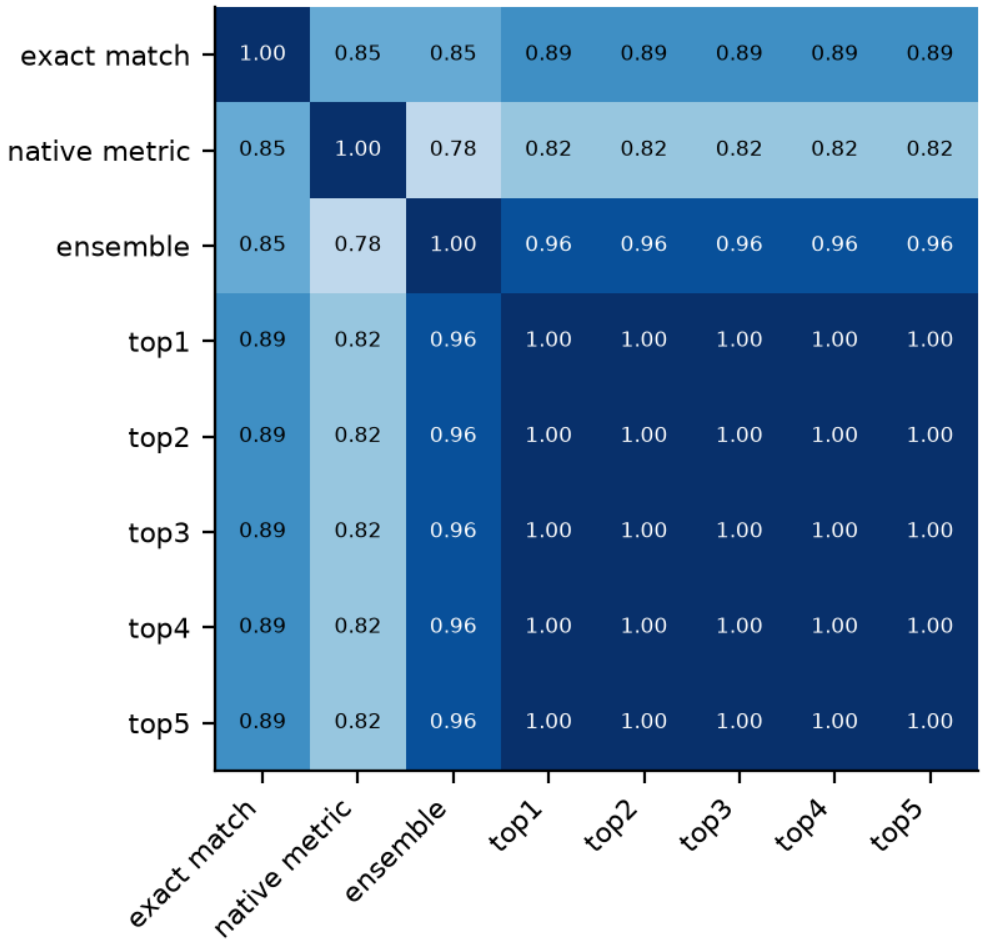


Figure 37: Kendall τ between the Bradley–Terry rankings of each pair of rules.

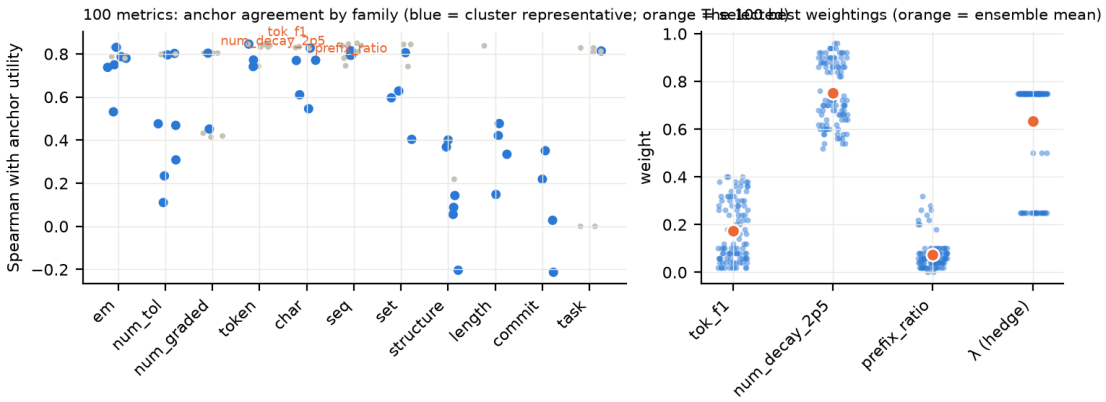


Figure 38: Metric selection: anchor agreement of every library metric with the selected set marked.

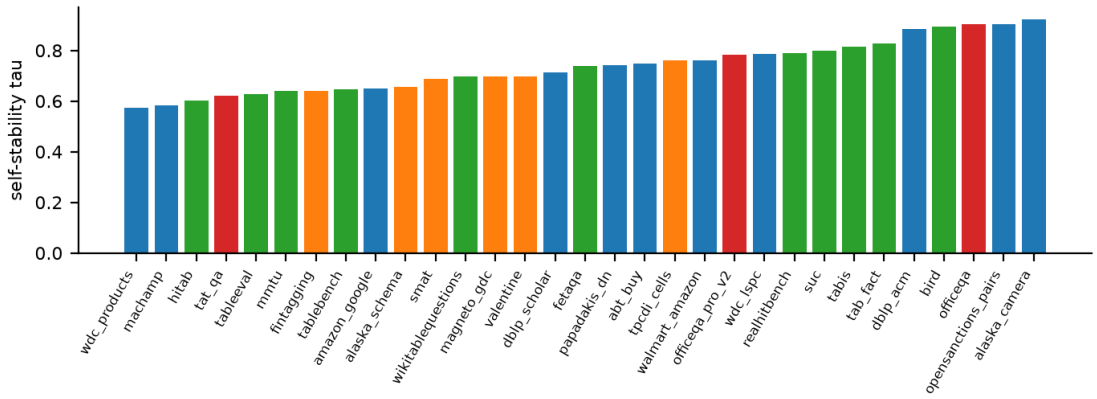


Figure 39: Self-stability of each dataset's model ranking (Kendall τ between bootstrap halves).

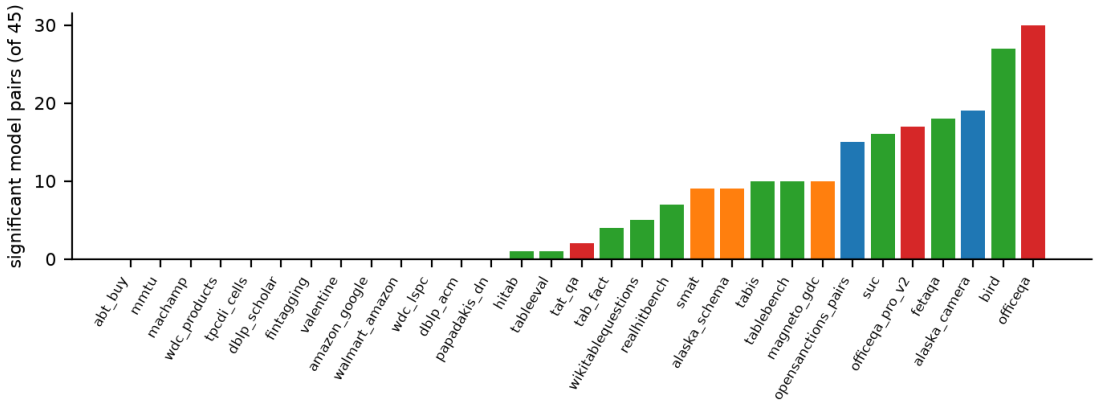


Figure 40: How many of the 45 model pairs each dataset separates at the item level under the headline rule (colour = family).

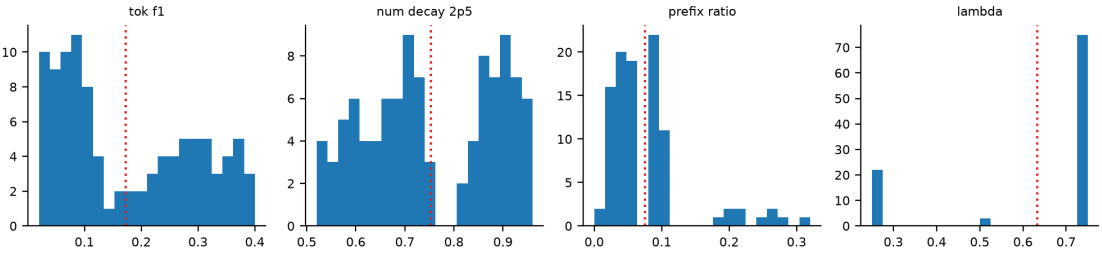


Figure 41: Weights and λ of the 100 best grid points that form the ensemble (dotted = mean).

B All tables

Every saved table, one file each under `tables/`; include any in the main text with `\input`.

Table 8: Baselines under the same objective.

baseline	rho	auc	js	bad	tau_native	J
native_style_metric	0.826	0.935	0.031	0.014	0.992	0.925
em_norm	0.849	0.963	0.024	0.012	0.916	0.927
tok_f1	0.859	0.965	0.079	0.032	0.884	0.906
rouge_l	0.862	0.965	0.082	0.032	0.884	0.906
num_tol_1	0.806	0.919	0.032	0.016	0.942	0.909
uniform_over_selected_no_gate	0.882	0.970	0.098	0.028	0.907	0.912
ensemble_top100	0.904	0.996	0.054	0.032	0.893	0.933

Table 9: Sampled designs vs the exhaustive grid (best $J = 0.9337$ over 6630 points).

design	best_J_mean	best_J_sd	regret_mean	regret_max	min_pairwise_L1	discrepancy	zero_share
dirichlet_lattice	0.9333	0.0004	0.0004	0.0010	0.0400	0.2728	0.0615
lhs_simplex	0.9331	0.0003	0.0006	0.0011	0.0400	0.2702	0.0555
uniform_lattice	0.9332	0.0004	0.0005	0.0015	0.0400	0.2639	0.1115

Table 10: Dataset diagnostics under the headline rule.

dataset	family	mean_score	spread_ across_models	floor	ceiling	share_ constant_ bootstrap_ rankings	ranking_self_ stability_tau	tau_to_ others_mean	n_items	exclude_ from_ objective
abt_buy	EM	0.971	0.060	no	no	0.000	0.748	0.353	100	no
alaska_camera	EM	0.945	0.190	no	no	0.000	0.924	-0.200	100	no
alaska_schema	SM	0.911	0.168	no	no	0.000	0.655	0.001	93	no
amazon_google	EM	0.935	0.100	no	no	0.000	0.649	-0.011	100	no
bird	TQA	0.255	0.282	no	no	0.000	0.893	0.210	100	no
dblp_acm	EM	0.987	0.090	no	no	0.000	0.883	0.309	100	no
dblp_scholar	EM	0.948	0.060	no	no	0.000	0.713	-0.020	100	no
fetaqa	TQA	0.555	0.411	no	no	0.000	0.740	0.197	100	no
fintagging	SM	0.830	0.076	no	no	0.000	0.639	0.298	100	no
hitab	TQA	0.776	0.071	no	no	0.000	0.601	-0.147	100	no
machamp	EM	0.887	0.070	no	no	0.000	0.584	0.380	100	no
magneto_gdc	SM	0.919	0.315	no	no	0.000	0.697	0.246	100	no
mmtu	TQA	0.867	0.086	no	no	0.000	0.639	0.263	100	no
officeqa	FIN	0.286	0.312	no	no	0.000	0.903	0.327	100	no
officeqa_pro_v2	FIN	0.312	0.284	no	no	0.000	0.782	0.198	90	no
opensanctions_pairs	EM	0.835	0.240	no	no	0.000	0.905	0.037	100	no
papadakis_dn	EM	0.940	0.030	no	no	0.002	0.741	0.023	100	no
realhitbench	TQA	0.721	0.150	no	no	0.000	0.789	0.347	100	no
smat	SM	0.889	0.212	no	no	0.000	0.687	0.172	100	no
suc	TQA	0.952	0.165	no	no	0.000	0.800	0.314	100	no
tab_fact	TQA	0.802	0.200	no	no	0.000	0.827	0.000	100	no
tabis	TQA	0.879	0.195	no	no	0.000	0.814	0.237	100	no
tablebench	TQA	0.860	0.173	no	no	0.000	0.646	0.215	100	no
tableeval	TQA	0.775	0.112	no	no	0.000	0.629	0.144	100	no
tat_qa	FIN	0.798	0.076	no	no	0.000	0.623	0.338	100	no
tpcdi_cells	SM	0.417	0.020	no	no	0.000	0.761	0.311	100	no
valentine	SM	0.983	0.035	no	yes	0.000	0.697	0.242	100	yes
walmart_amazon	EM	0.959	0.060	no	no	0.000	0.761	0.349	100	no
wdc_lspc	EM	0.982	0.040	no	yes	0.004	0.788	0.262	100	yes
wdc_products	EM	0.978	0.020	no	yes	0.006	0.573	0.184	100	yes
wikitablequestions	TQA	0.851	0.125	no	no	0.000	0.696	0.351	100	no

Table 11: Ensemble summary.

quantity	value
selected	["tok_f1", "num_decay_2p5", "prefix_ratio"]
selected_idx	[33, 24, 51]
hedge_col	89
alpha	"rho": 0.3, "auc": 0.15, "consistency": 0.2, "native": 0.15, "bad": 0.2
grid_step	0.020
n_grid	6630
top_k	100
mean_ weights	"tok_f1": 0.17259999999999998, "num_decay_2p5": 0.7523999999999998, "prefix_ratio": 0.075
mean_ lambda	0.632
weight_ spread	"tok_f1": [0.02, 0.4], "num_decay_2p5": [0.52, 0.96], "prefix_ratio": [0.0, 0.32]
J_top1	0.934
J_top100	0.932
J_median_ grid	0.924
terms_top1	"rho": 0.9014035670162496, "auc": 0.9964055948226619, "js": 0.04391693277745681, "tau_native": 0.8875778974669173, "bad": 0.033203903955460724
terms_ ensemble_ mean	"rho": 0.9040260890464176, "auc": 0.9964741602775785, "js": 0.0540873623390991, "tau_native": 0.8927985765748007, "bad": 0.03218762943625826

Table 12: Mean ensemble spread per dataset.

dataset	spread
abt_buy	0.000
wdc_lspc	0.000
walmart_amazon	0.000
tab_fact	0.000
papadakis_dn	0.000
machamp	0.000
opensanctions_pairs	0.000
alaska_camera	0.000
dblp_acm	0.000
dblp_scholar	0.000
amazon_google	0.000
wdc_products	0.000
valentine	0.005
magneto_gdc	0.015
alaska_schema	0.019
suc	0.020
smat	0.028
wikitablequestions	0.028
tableeval	0.036
tablebench	0.041
mmtu	0.046
hitab	0.050
fintagging	0.057
tat_qa	0.063
realhitbench	0.072
tpcdi_cells	0.075
bird	0.085
officeqa_pro_v2	0.137
officeqa	0.149
tabis	0.169
fetaqa	0.241

Table 13: The 25 best weightings with every objective term.

rank	rho	auc	js	bad	tau__native	J	w__tok__f1	w__num__decay__2p5	w__prefix__ratio	lambda
1	0.901	0.996	0.044	0.033	0.888	0.934	0.080	0.900	0.020	0.250
2	0.900	0.996	0.044	0.033	0.889	0.934	0.040	0.900	0.060	0.250
3	0.901	0.996	0.044	0.033	0.888	0.934	0.060	0.900	0.040	0.250
4	0.901	0.996	0.044	0.033	0.888	0.934	0.060	0.920	0.020	0.250
5	0.900	0.996	0.043	0.033	0.886	0.934	0.060	0.900	0.040	0.750
6	0.900	0.996	0.044	0.032	0.889	0.933	0.020	0.900	0.080	0.250
7	0.900	0.996	0.043	0.033	0.888	0.933	0.040	0.920	0.040	0.250
8	0.901	0.996	0.043	0.033	0.885	0.933	0.040	0.940	0.020	0.250
9	0.900	0.997	0.043	0.033	0.886	0.933	0.080	0.900	0.020	0.750
10	0.899	0.996	0.042	0.032	0.886	0.933	0.020	0.900	0.080	0.750
11	0.900	0.996	0.042	0.033	0.885	0.933	0.040	0.900	0.060	0.750
12	0.899	0.996	0.042	0.033	0.885	0.933	0.020	0.960	0.020	0.250
13	0.906	0.997	0.057	0.032	0.897	0.933	0.220	0.700	0.080	0.750
14	0.907	0.997	0.060	0.032	0.899	0.933	0.280	0.640	0.080	0.750
15	0.906	0.997	0.057	0.032	0.897	0.933	0.200	0.700	0.100	0.750
16	0.902	0.996	0.048	0.033	0.889	0.933	0.080	0.880	0.040	0.250
17	0.908	0.997	0.061	0.032	0.899	0.933	0.280	0.620	0.100	0.750
18	0.899	0.996	0.043	0.033	0.885	0.933	0.020	0.940	0.040	0.250
19	0.909	0.997	0.063	0.032	0.902	0.933	0.400	0.540	0.060	0.750
20	0.899	0.996	0.042	0.033	0.883	0.933	0.040	0.920	0.040	0.750
21	0.907	0.997	0.058	0.032	0.897	0.933	0.260	0.660	0.080	0.750
22	0.908	0.997	0.061	0.032	0.899	0.933	0.300	0.620	0.080	0.750
23	0.905	0.997	0.056	0.032	0.896	0.933	0.240	0.700	0.060	0.750
24	0.906	0.997	0.058	0.032	0.897	0.933	0.280	0.660	0.060	0.750
25	0.905	0.997	0.056	0.032	0.896	0.933	0.200	0.720	0.080	0.750

Table 14: Full leaderboard under rule exact match; weights: –.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.045	0.000	1.000	1.000	3.000	0.789	1.000	1.000	2.000	0.857	1.000	1.000	1.000	1.000	1.000	0.768
Gemini 3.8 Flash	0.036	0.015	2.000	1.000	4.000	0.109	3.000	2.000	5.000	0.012	3.000	3.000	3.000	3.000	3.000	0.763
GPT-5.5	0.035	0.015	3.000	1.000	4.000	0.101	2.000	1.000	4.000	0.127	2.000	2.000	2.000	2.000	2.000	0.763
DeepSeek-R1	0.022	0.015	4.000	3.000	7.000	0.001	4.000	2.988	7.000	0.000	4.000	4.000	4.000	4.000	5.000	0.756
Gemini 3.1 Pro	0.019	0.015	5.000	3.000	8.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.754
Grok 4	0.015	0.015	6.000	4.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	7.000	6.000	7.000	0.753
Claude Sonnet 5	0.012	0.015	7.000	4.000	8.000	0.000	5.000	3.000	7.000	0.000	5.000	5.000	5.000	5.000	4.000	0.751
GPT-5.4-mini	0.007	0.015	8.000	5.000	8.000	0.000	9.000	7.000	9.000	0.000	8.000	8.000	8.000	9.000	9.000	0.749
DeepSeek-chat	-0.020	0.015	9.000	9.000	10.000	0.000	8.000	7.000	9.000	0.000	9.000	9.000	9.000	8.000	8.000	0.735
gpt-oss-120b	-0.037	0.015	10.000	9.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.727
Llama 3.3 70B	-0.134	0.015	11.000	11.000	11.000	0.000	11.000	11.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.678

Table 15: Full leaderboard under rule ensemble; weights: tok_f1 0.17, num_decay_2p5 0.75, prefix_ratio 0.07, lambda 0.63.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.118	0.015	1.000	1.000	2.000	0.742	3.000	1.000	4.000	0.058	3.000	3.000	3.000	3.000	2.000	0.834
Claude Opus 5	0.111	0.000	2.000	1.000	2.000	0.258	1.000	1.000	3.000	0.713	1.000	1.000	1.000	1.000	1.000	0.834
GPT-5.5	0.034	0.015	3.000	3.000	6.000	0.000	2.000	1.000	3.000	0.220	2.000	2.000	2.000	2.000	3.000	0.823
Claude Sonnet 5	0.029	0.015	4.000	3.000	7.000	0.000	4.000	4.000	7.000	0.000	5.000	5.000	4.000	4.000	4.000	0.815
DeepSeek-R1	0.029	0.015	5.000	3.000	7.000	0.000	5.000	3.000	7.000	0.000	4.000	5.000	5.000	5.000	5.000	0.815
Gemini 3.1 Pro	0.022	0.015	6.000	3.000	7.000	0.000	6.000	4.000	8.000	0.000	7.000	7.000	6.000	6.000	6.000	0.811
Grok 4	0.012	0.015	7.000	4.000	7.000	0.000	7.000	4.000	8.000	0.000	6.000	5.000	7.000	7.000	8.000	0.810
GPT-5.4-mini	-0.024	0.015	8.000	8.000	9.000	0.000	8.000	6.000	9.000	0.000	8.000	8.000	8.000	8.000	7.000	0.804
DeepSeek-chat	-0.035	0.015	9.000	8.000	9.000	0.000	9.000	7.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.797
gpt-oss-120b	-0.103	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.783
Llama 3.3 70B	-0.193	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.747

Table 16: Leaderboard under the headline rule (ensemble).

rank	model	BT strength	SE	Borda rank	Borda 95%	P(first) BT	mean score
1	Gemini 3.8 Flash	0.118	0.015	3	[1, 4]	0.740	0.834
2	Claude Opus 5	0.111	0.000	1	[1, 3]	0.260	0.834
3	GPT-5.5	0.034	0.015	2	[1, 3]	0.000	0.823
4	Claude Sonnet 5	0.029	0.015	4	[4, 7]	0.000	0.815
5	DeepSeek-R1	0.029	0.015	5	[3, 7]	0.000	0.815
6	Gemini 3.1 Pro	0.022	0.015	6	[4, 8]	0.000	0.811
7	Grok 4	0.012	0.015	7	[4, 8]	0.000	0.810
8	GPT-5.4-mini	-0.024	0.015	8	[6, 9]	0.000	0.804
9	DeepSeek-chat	-0.035	0.015	9	[7, 9]	0.000	0.797
10	gpt-oss-120b	-0.103	0.015	10	[10, 11]	0.000	0.783
11	Llama 3.3 70B	-0.193	0.015	11	[10, 11]	0.000	0.747

Table 17: Full leaderboard under rule native metric; weights: –.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.059	0.015	1.000	1.000	2.000	0.708	3.000	1.000	4.000	0.036	3.000	3.000	3.000	3.000	3.000	0.793
GPT-5.5	0.054	0.015	2.000	1.000	3.000	0.279	1.000	1.000	3.000	0.465	2.000	2.000	2.000	1.000	2.000	0.792
Claude Opus 5	0.041	0.000	3.000	2.000	5.000	0.011	2.000	1.000	3.000	0.491	1.000	1.000	1.000	2.000	1.000	0.793
Gemini 3.1 Pro	0.034	0.015	4.000	3.000	6.000	0.002	6.000	4.000	7.000	0.000	6.000	5.500	5.000	6.000	6.000	0.781
DeepSeek-R1	0.026	0.015	5.000	3.000	7.000	0.000	4.000	3.000	7.000	0.000	4.000	7.000	4.000	5.000	5.000	0.781
Grok 4	0.025	0.015	6.000	4.000	7.000	0.000	7.000	4.000	8.000	0.000	7.000	5.500	7.000	7.000	7.000	0.779
Claude Sonnet 5	0.010	0.015	7.000	6.000	8.000	0.000	5.000	4.000	7.000	0.000	5.000	4.000	6.000	4.000	4.000	0.776
DeepSeek-chat	-0.006	0.015	8.000	7.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.761
GPT-5.4-mini	-0.016	0.015	9.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	8.000	8.000	8.000	0.770
gpt-oss-120b	-0.059	0.015	10.000	10.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.748
Llama 3.3 70B	-0.169	0.015	11.000	11.000	11.000	0.000	11.000	11.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.699

Table 18: Full leaderboard under rule top1; weights: tok_f1 0.08, num_decay_2p5 0.90, prefix_ratio 0.02, lambda 0.25.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.117	0.015	1.000	1.000	2.000	0.777	3.000	1.000	4.000	0.059	3.000	3.000	3.000	3.000	2.000	0.841
Claude Opus 5	0.109	0.000	2.000	1.000	2.000	0.223	1.000	1.000	3.000	0.711	1.000	1.000	1.000	1.000	1.000	0.840
GPT-5.5	0.034	0.015	3.000	3.000	6.000	0.000	2.000	1.000	3.000	0.222	2.000	2.000	2.000	2.000	3.000	0.828
DeepSeek-R1	0.033	0.015	4.000	3.000	7.000	0.000	5.000	3.000	7.000	0.000	4.000	4.000	5.000	5.000	5.000	0.822
Claude Sonnet 5	0.028	0.015	5.000	3.000	7.000	0.000	4.000	4.000	7.000	0.000	5.000	5.000	4.000	4.000	4.000	0.821
Gemini 3.1 Pro	0.021	0.015	6.000	3.000	7.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.816
Grok 4	0.012	0.015	7.000	4.000	7.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	7.000	6.000	8.000	0.815
GPT-5.4-mini	-0.024	0.015	8.000	8.000	9.000	0.000	8.000	6.000	9.000	0.000	8.000	8.000	8.000	8.000	7.000	0.809
DeepSeek-chat	-0.035	0.015	9.000	8.000	9.000	0.000	9.000	7.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.802
gpt-oss-120b	-0.104	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.788
Llama 3.3 70B	-0.192	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.751

Table 19: Full leaderboard under rule top2; weights: tok_f1 0.04, num_decay_2p5 0.90, prefix_ratio 0.06, lambda 0.25.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.015	1.000	1.000	2.000	0.715	3.000	1.000	4.000	0.057	3.000	3.000	3.000	3.000	2.000	0.839
Claude Opus 5	0.109	0.000	2.000	1.000	2.000	0.285	1.000	1.000	3.000	0.740	1.000	1.000	1.000	1.000	1.000	0.839
GPT-5.5	0.035	0.015	3.000	3.000	6.000	0.000	2.000	1.000	3.000	0.197	2.000	2.000	2.000	2.000	3.000	0.827
DeepSeek-R1	0.032	0.015	4.000	3.000	7.000	0.000	4.000	3.000	7.000	0.000	4.000	4.000	5.000	4.000	5.000	0.821
Claude Sonnet 5	0.027	0.015	5.000	3.000	7.000	0.000	5.000	3.988	7.000	0.000	5.000	5.000	4.000	5.000	4.000	0.820
Gemini 3.1 Pro	0.021	0.015	6.000	3.000	7.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.815
Grok 4	0.011	0.015	7.000	4.000	7.000	0.000	6.000	4.000	8.025	0.000	6.000	6.000	7.000	6.000	8.000	0.814
GPT-5.4-mini	-0.025	0.015	8.000	8.000	9.000	0.000	8.000	6.000	9.000	0.000	8.000	8.000	8.000	8.000	7.000	0.808
DeepSeek-chat	-0.036	0.015	9.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.800
gpt-oss-120b	-0.097	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.787
Llama 3.3 70B	-0.193	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.750

Table 20: Full leaderboard under rule top3; weights: tok_f1 0.06, num_decay_2p5 0.90, prefix_ratio 0.04, lambda 0.25.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.015	1.000	1.000	2.000	0.673	3.000	1.000	4.000	0.058	3.000	3.000	3.000	3.000	2.000	0.840
Claude Opus 5	0.111	0.000	2.000	1.000	2.000	0.327	1.000	1.000	3.000	0.721	1.000	1.000	1.000	1.000	1.000	0.840
GPT-5.5	0.034	0.015	3.000	3.000	6.000	0.000	2.000	1.000	3.000	0.217	2.000	2.000	2.000	2.000	3.000	0.828
DeepSeek-R1	0.032	0.015	4.000	3.000	7.000	0.000	5.000	3.000	7.000	0.000	4.000	4.000	5.000	5.000	5.000	0.821
Claude Sonnet 5	0.028	0.015	5.000	3.000	7.000	0.000	4.000	4.000	7.000	0.000	5.000	5.000	4.000	4.000	4.000	0.820
Gemini 3.1 Pro	0.022	0.015	6.000	3.000	7.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.816
Grok 4	0.012	0.015	7.000	4.000	7.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	7.000	6.000	8.000	0.814
GPT-5.4-mini	-0.024	0.015	8.000	8.000	9.000	0.000	8.000	6.000	9.000	0.000	8.000	8.000	8.000	8.000	7.000	0.808
DeepSeek-chat	-0.035	0.015	9.000	8.000	9.000	0.000	9.000	7.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.801
gpt-oss-120b	-0.101	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.787
Llama 3.3 70B	-0.193	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.750

Table 21: Full leaderboard under rule top4; weights: tok_f1 0.06, num_decay_2p5 0.92, prefix_ratio 0.02, lambda 0.25.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.116	0.015	1.000	1.000	2.000	0.721	3.000	1.000	4.000	0.058	3.000	3.000	3.000	3.000	2.000	0.841
Claude Opus 5	0.109	0.000	2.000	1.000	2.000	0.279	1.000	1.000	3.000	0.714	1.000	1.000	1.000	1.000	1.000	0.841
GPT-5.5	0.034	0.015	3.000	3.000	6.000	0.000	2.000	1.000	3.000	0.218	2.000	2.000	2.000	2.000	3.000	0.829
DeepSeek-R1	0.032	0.015	4.000	3.000	7.000	0.000	5.000	3.000	7.000	0.000	4.000	4.000	5.000	5.000	5.000	0.822
Claude Sonnet 5	0.029	0.015	5.000	3.000	7.000	0.000	4.000	4.000	7.000	0.000	5.000	5.000	4.000	4.000	4.000	0.822
Gemini 3.1 Pro	0.021	0.015	6.000	3.000	7.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.817
Grok 4	0.013	0.015	7.000	4.000	7.000	0.000	6.000	4.000	9.000	0.000	6.000	6.000	7.000	6.000	8.000	0.816
GPT-5.4-mini	-0.024	0.015	8.000	8.000	9.000	0.000	8.000	5.988	9.000	0.000	8.000	8.000	8.000	8.000	7.000	0.809
DeepSeek-chat	-0.035	0.015	9.000	8.000	9.000	0.000	9.000	7.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.802
gpt-oss-120b	-0.103	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.788
Llama 3.3 70B	-0.192	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.751

Table 22: Full leaderboard under rule top5; weights: tok_f1 0.06, num_decay_2p5 0.90, prefix_ratio 0.04, lambda 0.75.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.116	0.015	1.000	1.000	2.000	0.695	3.000	1.000	4.000	0.055	3.000	3.000	3.000	3.000	2.000	0.839
Claude Opus 5	0.111	0.000	2.000	1.000	2.000	0.305	1.000	1.000	3.000	0.716	1.000	1.000	1.000	1.000	1.000	0.839
GPT-5.5	0.034	0.015	3.000	3.000	6.000	0.000	2.000	1.000	3.000	0.222	2.000	2.000	2.000	2.000	3.000	0.827
DeepSeek-R1	0.029	0.015	4.000	3.000	7.000	0.000	5.000	3.000	7.000	0.000	4.000	5.000	5.000	5.000	5.000	0.819
Claude Sonnet 5	0.029	0.015	5.000	3.000	7.000	0.000	4.000	4.000	7.000	0.000	5.000	5.000	4.000	4.000	4.000	0.820
Gemini 3.1 Pro	0.018	0.015	6.000	3.000	7.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.815
Grok 4	0.013	0.015	7.000	4.000	7.000	0.000	6.000	4.000	9.000	0.000	6.000	5.000	7.000	6.000	8.000	0.814
GPT-5.4-mini	-0.023	0.015	8.000	8.000	9.000	0.000	8.000	5.000	9.000	0.000	8.000	8.000	8.000	8.000	7.000	0.808
DeepSeek-chat	-0.034	0.015	9.000	8.000	9.000	0.000	9.000	7.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.800
gpt-oss-120b	-0.101	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.787
Llama 3.3 70B	-0.192	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.749

Table 23: Leave-one-dataset-out.

held_out	selected_lodo	rho_ heldout_ lodo	rho_ heldout_ full	js_all_ lodo	same_ selection
abt_buy	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.919	0.919	0.058	yes
alaska_camera	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.921	0.921	0.058	yes
alaska_schema	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.927	0.928	0.058	yes
amazon_google	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.918	0.918	0.058	yes
bird	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.877	0.878	0.056	yes
dblp_acm	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.919	0.919	0.058	yes
dblp_scholar	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.920	0.920	0.058	yes
fetaqa	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.874	0.888	0.061	yes
fintagging	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.907	0.906	0.072	yes
hitab	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.887	0.888	0.057	yes
machamp	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.921	0.921	0.058	yes
magneto_gdc	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.938	0.939	0.058	yes
mmtu	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.893	0.893	0.058	yes
officeqa	['tok_f1', 'num_tol_10', 'prefix_ratio']	0.890	0.900	0.038	no
officeqa_pro_ v2	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.903	0.905	0.057	yes
opensanctions_ pairs	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.919	0.919	0.058	yes
papadakis_dn	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.910	0.910	0.058	yes
realhitbench	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.896	0.897	0.057	yes
smat	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.941	0.941	0.058	yes
suc	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.865	0.867	0.057	yes
tab_fact	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.918	0.918	0.058	yes
tabis	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.598	0.598	0.083	yes
tablebench	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.884	0.885	0.057	yes
tableeval	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.869	0.870	0.057	yes
tat_qa	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.896	0.896	0.058	yes
tpcdi_cells	['tok_f1', 'num_decay_2p5', 'prefix_ratio', 'em_raw']	0.867	0.868	0.066	no
valentine	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.899	0.902	0.058	yes
walmart_ amazon	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.917	0.917	0.058	yes
wdc_lspc	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.921	0.921	0.058	yes
wdc_products	['tok_f1', 'num_decay_2p5', 'prefix_ratio']	0.921	0.921	0.058	yes
wikitablequestions	['tok_f1', 'num_decay_2p5', 'prefix_ratio', 'em_raw']	0.876	0.867	0.065	no

Table 24: Every library metric: anchor ρ, AUC, gap, family, cluster.

metric	rho	auc	gap	family	representative	cluster
rouge_l	0.849	0.965	0.800	seq	no	tok_f1
rouge1	0.846	0.965	0.802	seq	no	tok_f1
tok_f1	0.846	0.965	0.802	token	yes	tok_f1
stem_f1	0.846	0.965	0.802	token	no	tok_f1
tok_f0p5	0.845	0.965	0.828	token	no	tok_f1
jaccard	0.844	0.964	0.836	set	no	tok_f1
dice	0.844	0.964	0.794	set	no	tok_f1
content_f1	0.843	0.963	0.786	token	no	tok_f1
tok_f2	0.841	0.965	0.749	token	no	tok_f1
tok_prec	0.839	0.965	0.840	token	no	tok_f1
ter_score	0.839	0.966	0.912	seq	no	em_norm
bigram_f1	0.838	0.966	0.871	token	no	tok_f1
rouge2	0.838	0.966	0.871	seq	no	tok_f1
extra_token_score	0.838	0.964	0.838	length	no	tok_f1
jaro_winkler	0.834	0.968	0.490	char	no	lev_sim
jaro_sim	0.833	0.968	0.503	char	no	lev_sim
trigram_f1	0.832	0.966	0.897	token	no	em_norm
bigram_prec	0.832	0.966	0.890	token	no	tok_f1
em_no_space	0.831	0.963	0.925	em	no	em_norm
em_norm	0.831	0.963	0.925	em	yes	em_norm
lcs_substr_ratio	0.829	0.970	0.747	char	no	lev_sim
bool_label_agree	0.828	0.960	0.921	task	no	em_norm
attribute_match_score	0.828	0.960	0.921	task	no	em_norm
lev_sim	0.827	0.970	0.725	char	yes	lev_sim
damerau_sim	0.826	0.970	0.724	char	no	lev_sim
bleu1	0.814	0.953	0.561	seq	yes	bleu1
native_style_metric	0.814	0.935	0.866	task	yes	native_style_metric
year_aware_em	0.812	0.953	0.907	task	no	em_norm
wer_score	0.810	0.971	0.827	seq	no	lev_sim
numeric_or_em	0.809	0.932	0.864	task	no	native_style_metric
pred_in_gold	0.806	0.952	0.903	set	yes	pred_in_gold
num_decay_10	0.803	0.927	0.825	num_graded	no	num_decay_2p5
num_decay_5	0.803	0.927	0.812	num_graded	no	num_decay_2p5
num_decay_0p5	0.803	0.927	0.674	num_graded	no	num_decay_2p5
num_decay_1	0.803	0.927	0.728	num_graded	no	num_decay_2p5
num_decay_2p5	0.803	0.927	0.787	num_graded	yes	num_decay_2p5
num_tol_5	0.803	0.915	0.829	num_tol	no	num_tol_1
num_tol_10	0.802	0.913	0.825	num_tol	yes	num_tol_10
num_tol_0p1	0.800	0.923	0.846	num_tol	no	num_tol_1
num_tol_2	0.798	0.917	0.835	num_tol	no	num_tol_1
num_tol_20	0.798	0.907	0.814	num_tol	no	num_tol_10
num_tol_0p5	0.797	0.920	0.841	num_tol	no	num_tol_1
num_tol_1	0.795	0.919	0.838	num_tol	yes	num_tol_1
meteor_lite	0.794	0.888	0.467	seq	yes	meteor_lite
bleu2	0.794	0.956	0.392	seq	yes	bleu2
em_list_ordered	0.790	0.935	0.870	em	no	em_list_set
em_casefold	0.788	0.902	0.804	em	no	em_raw
em_list_set	0.787	0.934	0.868	em	yes	em_list_set
bleu4	0.780	0.953	0.278	seq	no	bleu2
em_raw	0.780	0.888	0.775	em	yes	em_raw
list_precision	0.780	0.929	0.834	em	no	list_recall
bigram_rec	0.772	0.899	0.792	token	yes	bigram_rec
chrf2	0.771	0.922	0.630	char	yes	chrf2
prefix_ratio	0.769	0.866	0.700	char	yes	prefix_ratio
em_first_token	0.750	0.866	0.732	em	yes	em_first_token
rouge_l_recall	0.745	0.838	0.664	seq	no	tok_rec
content_rec	0.743	0.838	0.665	token	no	tok_rec
overlap_coef	0.741	0.839	0.666	set	no	tok_rec
tok_rec	0.741	0.838	0.666	token	yes	tok_rec
list_recall	0.738	0.888	0.761	em	yes	list_recall
gold_in_pred	0.628	0.779	0.558	set	yes	gold_in_pred
chrf4	0.610	0.654	0.297	char	yes	chrf4
number_set_jaccard	0.596	0.734	0.423	set	yes	number_set_jaccard
chrf3	0.546	0.759	0.438	char	yes	chrf3
em_numbers_only	0.531	0.656	0.312	em	yes	em_numbers_only
not_abstaining	0.477	0.614	0.228	length	yes	not_abstaining
num_abs_0p01	0.476	0.620	0.240	num_tol	yes	num_abs_0p01
num_round_to_gold_precision	0.468	0.616	0.232	num_tol	yes	num_round_to_gold_precision
num_linear	0.451	0.569	0.167	num_graded	yes	num_linear
num_log_ratio	0.432	0.549	0.155	num_graded	no	num_sign_agree
length_ratio_score	0.421	0.726	0.305	length	yes	length_ratio_score
num_smape	0.419	0.546	0.134	num_graded	no	num_sign_agree
num_inv_rel	0.414	0.530	0.103	num_graded	no	num_sign_agree
entity_set_jaccard	0.403	0.624	0.234	set	yes	entity_set_jaccard
final_tag_present	0.401	0.619	0.238	structure	yes	final_tag_present
type_agree	0.368	0.650	0.300	structure	yes	type_agree
number_count_agree	0.351	0.624	0.249	commit	yes	number_count_agree
not_too_long	0.334	0.598	0.196	length	yes	not_too_long
num_magnitude_agree	0.308	0.541	0.082	num_tol	yes	num_magnitude_agree
num_sign_agree	0.234	0.507	0.014	num_tol	yes	num_sign_agree
single_candidate	0.219	0.667	0.335	structure	no	hedge_free
hedge_free	0.219	0.667	0.335	commit	yes	hedge_free
brevity_penalty	0.148	0.519	0.030	length	yes	brevity_penalty
list_length_match	0.143	0.574	0.149	structure	yes	list_length_match
num_unit_equiv_hit	0.110	0.507	0.015	num_tol	yes	num_unit_equiv_hit
unit_word_match	0.088	0.507	0.013	structure	yes	unit_word_match
percent_sign_agree	0.055	0.506	0.011	structure	yes	percent_sign_agree

first__last__number__agree	0.028	0.549	0.098	commit	yes	first__last__number__agree
emb__cos__bge	0.000	0.500	0.000	semantic	no	__constant
nli__pred__entails__gold	0.000	0.500	0.000	semantic	no	__constant
emb__cos__minilm	0.000	0.500	0.000	semantic	no	__constant
sql__result__match__placeholder	0.000	0.500	0.000	task	no	__constant
program__exec__match	0.000	0.500	0.000	task	no	__constant
bertscore__f__placeholder	0.000	0.500	0.000	semantic	no	__constant
emb__cos__minilm__thresh	0.000	0.500	0.000	semantic	no	__constant
emb__cos__bge__thresh	0.000	0.500	0.000	semantic	no	__constant
emb__question__free__placeholder	0.000	0.500	0.000	semantic	no	__constant
nli__gold__entails__pred	0.000	0.500	0.000	semantic	no	__constant
currency__symbol__agree	-0.204	0.463	-0.074	structure	yes	currency__symbol__agree
reasoning__final__consistent	-0.213	0.473	-0.055	commit	yes	reasoning__final__consistent

Table 25: Official-split estimates under the headline rule for every dataset and model.

dataset	model	suite_mean	official_ estimate	official_lo95	official_hi95	effective_n	suite_f1_pos	official_f1_ pos	official_f1_ lo95	official_f1_ hi95	official_ positive_ share
abt_buy	Claude Opus 5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	Claude Sonnet 5	0.970	0.993	0.985	1.000	63.075	0.968	0.968	0.925	1.000	0.108
abt_buy	DeepSeek-chat	0.960	0.976	0.933	0.998	63.075	0.958	0.893	0.735	0.990	0.108
abt_buy	DeepSeek-R1	0.970	0.978	0.931	1.000	63.075	0.969	0.904	0.755	1.000	0.108
abt_buy	Gemini 3.1 Pro	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.945	1.000	0.108
abt_buy	Gemini 3.8 Flash	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.944	1.000	0.108
abt_buy	GPT-5.4-mini	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.736	0.991	0.108
abt_buy	GPT-5.5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	gpt-oss-120b	0.940	0.987	0.974	0.996	63.075	0.935	0.935	0.871	0.978	0.108
abt_buy	Grok 4	0.960	0.976	0.935	0.998	63.075	0.958	0.893	0.739	0.990	0.108
abt_buy	Llama 3.3 70B	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.737	0.990	0.108
alaska_camera	Claude Opus 5	0.820	0.820	0.740	0.890	100.000	0.795	0.795	0.700	0.882	0.500
alaska_camera	Claude Sonnet 5	0.980	0.980	0.950	1.000	100.000	0.981	0.981	0.948	1.000	0.500
alaska_camera	DeepSeek-chat	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	DeepSeek-R1	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.968	1.000	0.500
alaska_camera	Gemini 3.1 Pro	0.810	0.810	0.730	0.880	100.000	0.782	0.782	0.667	0.869	0.500
alaska_camera	Gemini 3.8 Flash	0.890	0.890	0.830	0.950	100.000	0.884	0.884	0.804	0.945	0.500
alaska_camera	GPT-5.4-mini	0.970	0.970	0.930	1.000	100.000	0.971	0.971	0.932	1.000	0.500
alaska_camera	GPT-5.5	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	gpt-oss-120b	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	Grok 4	0.940	0.940	0.890	0.980	100.000	0.940	0.940	0.889	0.981	0.500
alaska_camera	Llama 3.3 70B	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.968	1.000	0.500
alaska_schema	Claude Opus 5	0.933	0.933	0.882	0.974	93.000	–	–	–	–	–
alaska_schema	Claude Sonnet 5	0.920	0.920	0.865	0.969	93.000	–	–	–	–	–
alaska_schema	DeepSeek-chat	0.935	0.935	0.884	0.981	93.000	–	–	–	–	–
alaska_schema	DeepSeek-R1	0.943	0.943	0.898	0.982	93.000	–	–	–	–	–
alaska_schema	Gemini 3.1 Pro	0.912	0.912	0.852	0.962	93.000	–	–	–	–	–
alaska_schema	Gemini 3.8 Flash	0.902	0.902	0.841	0.953	93.000	–	–	–	–	–
alaska_schema	GPT-5.4-mini	0.951	0.951	0.909	0.989	93.000	–	–	–	–	–
alaska_schema	GPT-5.5	0.912	0.912	0.854	0.963	93.000	–	–	–	–	–
alaska_schema	gpt-oss-120b	0.915	0.915	0.862	0.963	93.000	–	–	–	–	–
alaska_schema	Grok 4	0.912	0.912	0.853	0.964	93.000	–	–	–	–	–
alaska_schema	Llama 3.3 70B	0.783	0.783	0.701	0.857	93.000	–	–	–	–	–
amazon_google	Claude Opus 5	0.940	0.916	0.839	0.981	67.152	0.936	0.704	0.526	0.919	0.102
amazon_google	Claude Sonnet 5	0.960	0.935	0.868	0.984	67.152	0.957	0.758	0.588	0.935	0.102
amazon_google	DeepSeek-chat	0.960	0.949	0.893	0.998	67.152	0.957	0.796	0.614	0.990	0.102

amazon_google	DeepSeek-R1	0.944	0.931	0.865	0.983	67.152	0.939	0.740	0.571	0.927	0.102
amazon_google	Gemini 3.1 Pro	0.930	0.928	0.862	0.981	67.152	0.923	0.725	0.551	0.914	0.102
amazon_google	Gemini 3.8 Flash	0.930	0.928	0.859	0.981	67.152	0.923	0.725	0.548	0.913	0.102
amazon_google	GPT-5.4-mini	0.930	0.942	0.880	0.990	67.152	0.921	0.762	0.583	0.959	0.102
amazon_google	GPT-5.5	0.940	0.916	0.844	0.982	67.152	0.936	0.704	0.535	0.915	0.102
amazon_google	gpt-oss-120b	0.860	0.926	0.861	0.973	67.152	0.829	0.676	0.507	0.874	0.102
amazon_google	Grok 4	0.940	0.944	0.875	0.992	67.152	0.933	0.773	0.607	0.966	0.102
amazon_google	Llama 3.3 70B	0.950	0.932	0.861	0.982	67.152	0.946	0.747	0.561	0.941	0.102
bird	Claude Opus 5	0.389	0.408	0.315	0.509	74.082	–	–	–	–	–
bird	Claude Sonnet 5	0.329	0.336	0.248	0.436	74.082	–	–	–	–	–
bird	DeepSeek-chat	0.271	0.273	0.194	0.351	74.082	–	–	–	–	–
bird	DeepSeek-R1	0.290	0.321	0.231	0.409	74.082	–	–	–	–	–
bird	Gemini 3.1 Pro	0.213	0.252	0.167	0.341	74.082	–	–	–	–	–
bird	Gemini 3.8 Flash	0.404	0.421	0.333	0.514	74.082	–	–	–	–	–
bird	GPT-5.4-mini	0.231	0.267	0.178	0.360	74.082	–	–	–	–	–
bird	GPT-5.5	0.213	0.221	0.132	0.319	74.082	–	–	–	–	–
bird	gpt-oss-120b	0.121	0.139	0.072	0.214	74.082	–	–	–	–	–
bird	Grok 4	0.151	0.172	0.098	0.259	74.082	–	–	–	–	–
bird	Llama 3.3 70B	0.196	0.209	0.137	0.285	74.082	–	–	–	–	–
dblp_acm	Claude Opus 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Claude Sonnet 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-chat	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-R1	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.965	1.000	0.180
dblp_acm	Gemini 3.1 Pro	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Gemini 3.8 Flash	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	GPT-5.4-mini	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.965	1.000	0.180
dblp_acm	GPT-5.5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	gpt-oss-120b	0.910	0.967	0.945	0.987	72.166	0.899	0.899	0.829	0.960	0.180
dblp_acm	Grok 4	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Llama 3.3 70B	0.970	0.952	0.893	1.000	72.166	0.970	0.882	0.750	1.000	0.180
dblp_scholar	Claude Opus 5	0.950	0.956	0.901	0.996	70.463	0.950	0.888	0.752	0.981	0.186
dblp_scholar	Claude Sonnet 5	0.970	0.976	0.932	1.000	70.463	0.970	0.937	0.835	1.000	0.186
dblp_scholar	DeepSeek-chat	0.960	0.959	0.907	0.997	70.463	0.961	0.898	0.780	0.991	0.186
dblp_scholar	DeepSeek-R1	0.950	0.969	0.931	0.996	70.463	0.949	0.917	0.818	0.987	0.186
dblp_scholar	Gemini 3.1 Pro	0.940	0.965	0.927	0.993	70.463	0.939	0.906	0.812	0.980	0.186
dblp_scholar	Gemini 3.8 Flash	0.930	0.961	0.914	0.990	70.463	0.928	0.895	0.791	0.971	0.186
dblp_scholar	GPT-5.4-mini	0.940	0.965	0.925	0.992	70.463	0.939	0.906	0.799	0.980	0.186
dblp_scholar	GPT-5.5	0.960	0.959	0.905	0.996	70.463	0.961	0.898	0.784	0.991	0.186
dblp_scholar	gpt-oss-120b	0.910	0.967	0.943	0.986	70.463	0.903	0.903	0.829	0.961	0.186
dblp_scholar	Grok 4	0.950	0.969	0.928	0.996	70.463	0.949	0.917	0.812	0.988	0.186
dblp_scholar	Llama 3.3 70B	0.970	0.976	0.936	1.000	70.463	0.970	0.937	0.847	1.000	0.186

fetaqa	Claude Opus 5	0.626	0.626	0.556	0.687	100.000	–	–	–	–	–
fetaqa	Claude Sonnet 5	0.556	0.556	0.489	0.629	100.000	–	–	–	–	–
fetaqa	DeepSeek-chat	0.618	0.618	0.544	0.679	100.000	–	–	–	–	–
fetaqa	DeepSeek-R1	0.592	0.592	0.522	0.663	100.000	–	–	–	–	–
fetaqa	Gemini 3.1 Pro	0.607	0.607	0.536	0.673	100.000	–	–	–	–	–
fetaqa	Gemini 3.8 Flash	0.656	0.656	0.589	0.722	100.000	–	–	–	–	–
fetaqa	GPT-5.4-mini	0.340	0.340	0.276	0.406	100.000	–	–	–	–	–
fetaqa	GPT-5.5	0.617	0.617	0.549	0.682	100.000	–	–	–	–	–
fetaqa	gpt-oss-120b	0.610	0.610	0.538	0.678	100.000	–	–	–	–	–
fetaqa	Grok 4	0.643	0.643	0.573	0.708	100.000	–	–	–	–	–
fetaqa	Llama 3.3 70B	0.245	0.245	0.189	0.307	100.000	–	–	–	–	–
fintagging	Claude Opus 5	0.865	0.899	0.816	0.967	51.769	–	–	–	–	–
fintagging	Claude Sonnet 5	0.849	0.848	0.752	0.934	51.769	–	–	–	–	–
fintagging	DeepSeek-chat	0.789	0.775	0.666	0.871	51.769	–	–	–	–	–
fintagging	DeepSeek-R1	0.822	0.844	0.747	0.926	51.769	–	–	–	–	–
fintagging	Gemini 3.1 Pro	0.831	0.861	0.765	0.941	51.769	–	–	–	–	–
fintagging	Gemini 3.8 Flash	0.849	0.864	0.772	0.936	51.769	–	–	–	–	–
fintagging	GPT-5.4-mini	0.841	0.798	0.695	0.885	51.769	–	–	–	–	–
fintagging	GPT-5.5	0.847	0.861	0.775	0.939	51.769	–	–	–	–	–
fintagging	gpt-oss-120b	0.805	0.790	0.683	0.884	51.769	–	–	–	–	–
fintagging	Grok 4	0.839	0.828	0.735	0.919	51.769	–	–	–	–	–
fintagging	Llama 3.3 70B	0.789	0.757	0.647	0.867	51.769	–	–	–	–	–
hitab	Claude Opus 5	0.785	0.804	0.722	0.873	94.537	–	–	–	–	–
hitab	Claude Sonnet 5	0.776	0.779	0.696	0.851	94.537	–	–	–	–	–
hitab	DeepSeek-chat	0.816	0.834	0.762	0.896	94.537	–	–	–	–	–
hitab	DeepSeek-R1	0.789	0.793	0.720	0.865	94.537	–	–	–	–	–
hitab	Gemini 3.1 Pro	0.777	0.794	0.716	0.867	94.537	–	–	–	–	–
hitab	Gemini 3.8 Flash	0.757	0.775	0.697	0.847	94.537	–	–	–	–	–
hitab	GPT-5.4-mini	0.792	0.802	0.723	0.869	94.537	–	–	–	–	–
hitab	GPT-5.5	0.759	0.777	0.697	0.853	94.537	–	–	–	–	–
hitab	gpt-oss-120b	0.751	0.763	0.683	0.837	94.537	–	–	–	–	–
hitab	Grok 4	0.744	0.759	0.680	0.835	94.537	–	–	–	–	–
hitab	Llama 3.3 70B	0.795	0.804	0.729	0.873	94.537	–	–	–	–	–
machamp	Claude Opus 5	0.900	0.912	0.826	0.970	32.491	0.904	0.774	0.617	0.908	0.175
machamp	Claude Sonnet 5	0.900	0.923	0.852	0.975	32.491	0.902	0.793	0.642	0.924	0.175
machamp	DeepSeek-chat	0.890	0.892	0.801	0.962	32.491	0.897	0.742	0.604	0.895	0.175
machamp	DeepSeek-R1	0.900	0.897	0.808	0.963	32.491	0.906	0.751	0.594	0.895	0.175
machamp	Gemini 3.1 Pro	0.890	0.908	0.832	0.965	32.491	0.893	0.761	0.618	0.898	0.175
machamp	Gemini 3.8 Flash	0.900	0.912	0.836	0.973	32.491	0.904	0.774	0.608	0.920	0.175
machamp	GPT-5.4-mini	0.890	0.893	0.804	0.962	32.491	0.895	0.739	0.568	0.898	0.175

machamp	GPT-5.5	0.900	0.897	0.812	0.964	32.491	0.906	0.751	0.591	0.895	0.175
machamp	gpt-oss-120b	0.830	0.794	0.664	0.913	32.491	0.835	0.558	0.364	0.761	0.175
machamp	Grok 4	0.890	0.896	0.806	0.962	32.491	0.895	0.747	0.594	0.906	0.175
machamp	Llama 3.3 70B	0.870	0.880	0.791	0.947	32.491	0.874	0.698	0.529	0.846	0.175
magneto__gdc	Claude Opus 5	0.964	0.966	0.929	0.993	93.027	–	–	–	–	–
magneto__gdc	Claude Sonnet 5	0.945	0.945	0.895	0.983	93.027	–	–	–	–	–
magneto__gdc	DeepSeek-chat	0.937	0.940	0.889	0.981	93.027	–	–	–	–	–
magneto__gdc	DeepSeek-R1	0.973	0.976	0.948	1.000	93.027	–	–	–	–	–
magneto__gdc	Gemini 3.1 Pro	0.939	0.939	0.890	0.977	93.027	–	–	–	–	–
magneto__gdc	Gemini 3.8 Flash	0.954	0.958	0.919	0.991	93.027	–	–	–	–	–
magneto__gdc	GPT-5.4-mini	0.896	0.900	0.841	0.955	93.027	–	–	–	–	–
magneto__gdc	GPT-5.5	0.962	0.968	0.931	0.992	93.027	–	–	–	–	–
magneto__gdc	gpt-oss-120b	0.953	0.957	0.916	0.985	93.027	–	–	–	–	–
magneto__gdc	Grok 4	0.926	0.931	0.876	0.974	93.027	–	–	–	–	–
magneto__gdc	Llama 3.3 70B	0.658	0.663	0.572	0.747	93.027	–	–	–	–	–
mmtu	Claude Opus 5	0.883	0.833	0.753	0.920	86.538	–	–	–	–	–
mmtu	Claude Sonnet 5	0.883	0.834	0.750	0.919	86.538	–	–	–	–	–
mmtu	DeepSeek-chat	0.857	0.803	0.713	0.885	86.538	–	–	–	–	–
mmtu	DeepSeek-R1	0.835	0.775	0.679	0.862	86.538	–	–	–	–	–
mmtu	Gemini 3.1 Pro	0.896	0.852	0.764	0.930	86.538	–	–	–	–	–
mmtu	Gemini 3.8 Flash	0.891	0.845	0.764	0.924	86.538	–	–	–	–	–
mmtu	GPT-5.4-mini	0.879	0.835	0.752	0.919	86.538	–	–	–	–	–
mmtu	GPT-5.5	0.896	0.844	0.757	0.923	86.538	–	–	–	–	–
mmtu	gpt-oss-120b	0.822	0.753	0.656	0.852	86.538	–	–	–	–	–
mmtu	Grok 4	0.885	0.834	0.753	0.922	86.538	–	–	–	–	–
mmtu	Llama 3.3 70B	0.810	0.764	0.666	0.848	86.538	–	–	–	–	–
officeqa	Claude Opus 5	0.408	0.408	0.346	0.471	100.000	–	–	–	–	–
officeqa	Claude Sonnet 5	0.339	0.339	0.279	0.397	100.000	–	–	–	–	–
officeqa	DeepSeek-chat	0.124	0.124	0.080	0.173	100.000	–	–	–	–	–
officeqa	DeepSeek-R1	0.281	0.281	0.230	0.335	100.000	–	–	–	–	–
officeqa	Gemini 3.1 Pro	0.269	0.269	0.214	0.325	100.000	–	–	–	–	–
officeqa	Gemini 3.8 Flash	0.435	0.435	0.379	0.495	100.000	–	–	–	–	–
officeqa	GPT-5.4-mini	0.254	0.254	0.203	0.304	100.000	–	–	–	–	–
officeqa	GPT-5.5	0.358	0.358	0.301	0.420	100.000	–	–	–	–	–
officeqa	gpt-oss-120b	0.176	0.176	0.131	0.225	100.000	–	–	–	–	–
officeqa	Grok 4	0.273	0.273	0.216	0.332	100.000	–	–	–	–	–
officeqa	Llama 3.3 70B	0.225	0.225	0.181	0.270	100.000	–	–	–	–	–
officeqa__pro__v2	Claude Opus 5	0.502	0.502	0.430	0.574	90.000	–	–	–	–	–
officeqa__pro__v2	Claude Sonnet 5	0.322	0.322	0.257	0.386	90.000	–	–	–	–	–
officeqa__pro__v2	DeepSeek-chat	0.280	0.280	0.220	0.336	90.000	–	–	–	–	–
officeqa__pro__v2	DeepSeek-R1	0.256	0.256	0.188	0.326	90.000	–	–	–	–	–

officeqa_pro_v2	Gemini 3.1 Pro	0.267	0.267	0.207	0.333	90.000	–	–	–	–	–
officeqa_pro_v2	Gemini 3.8 Flash	0.413	0.413	0.349	0.484	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.4-mini	0.322	0.322	0.266	0.379	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.5	0.311	0.311	0.233	0.387	90.000	–	–	–	–	–
officeqa_pro_v2	gpt-oss-120b	0.218	0.218	0.156	0.281	90.000	–	–	–	–	–
officeqa_pro_v2	Grok 4	0.263	0.263	0.202	0.330	90.000	–	–	–	–	–
officeqa_pro_v2	Llama 3.3 70B	0.277	0.277	0.222	0.336	90.000	–	–	–	–	–
opensanctions_pairs	Claude Opus 5	0.940	0.906	0.831	0.969	76.249	0.935	0.935	0.878	0.979	0.769
opensanctions_pairs	Claude Sonnet 5	0.730	0.587	0.479	0.701	76.249	0.630	0.636	0.500	0.755	0.769
opensanctions_pairs	DeepSeek-chat	0.760	0.635	0.529	0.741	76.249	0.684	0.691	0.543	0.800	0.769
opensanctions_pairs	DeepSeek-R1	0.850	0.765	0.670	0.863	76.249	0.819	0.819	0.714	0.900	0.769
opensanctions_pairs	Gemini 3.1 Pro	0.970	0.964	0.917	1.000	76.249	0.969	0.976	0.943	1.000	0.769
opensanctions_pairs	Gemini 3.8 Flash	0.860	0.780	0.681	0.882	76.249	0.833	0.833	0.732	0.911	0.769
opensanctions_pairs	GPT-5.4-mini	0.900	0.865	0.781	0.940	76.249	0.891	0.905	0.833	0.960	0.769
opensanctions_pairs	GPT-5.5	0.750	0.608	0.503	0.724	76.249	0.658	0.658	0.525	0.778	0.769
opensanctions_pairs	gpt-oss-120b	0.780	0.666	0.561	0.780	76.249	0.718	0.725	0.588	0.815	0.769
opensanctions_pairs	Grok 4	0.850	0.776	0.676	0.868	76.249	0.824	0.830	0.734	0.913	0.769
opensanctions_pairs	Llama 3.3 70B	0.790	0.682	0.567	0.796	76.249	0.734	0.741	0.619	0.845	0.769
papadakis_dn	Claude Opus 5	0.920	0.955	0.894	0.995	32.601	0.789	0.322	0.131	0.814	0.018
papadakis_dn	Claude Sonnet 5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.559	0.939	0.018
papadakis_dn	DeepSeek-chat	0.940	0.993	0.986	0.998	32.601	0.833	0.757	0.534	0.923	0.018
papadakis_dn	DeepSeek-R1	0.940	0.993	0.986	0.998	32.601	0.833	0.757	0.527	0.918	0.018
papadakis_dn	Gemini 3.1 Pro	0.930	0.974	0.927	0.997	32.601	0.811	0.452	0.196	0.886	0.018
papadakis_dn	Gemini 3.8 Flash	0.940	0.992	0.984	0.998	32.601	0.824	0.702	0.442	0.893	0.018
papadakis_dn	GPT-5.4-mini	0.950	0.994	0.988	0.999	32.601	0.857	0.786	0.575	0.949	0.018
papadakis_dn	GPT-5.5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.564	0.944	0.018
papadakis_dn	gpt-oss-120b	0.930	0.992	0.984	0.998	32.601	0.788	0.682	0.431	0.890	0.018
papadakis_dn	Grok 4	0.950	0.994	0.988	0.999	32.601	0.857	0.786	0.562	0.951	0.018
papadakis_dn	Llama 3.3 70B	0.940	0.975	0.926	0.998	32.601	0.833	0.463	0.185	0.905	0.018
realhitbench	Claude Opus 5	0.777	0.778	0.702	0.845	98.080	–	–	–	–	–
realhitbench	Claude Sonnet 5	0.715	0.715	0.636	0.797	98.080	–	–	–	–	–
realhitbench	DeepSeek-chat	0.705	0.699	0.617	0.787	98.080	–	–	–	–	–
realhitbench	DeepSeek-R1	0.729	0.730	0.652	0.808	98.080	–	–	–	–	–
realhitbench	Gemini 3.1 Pro	0.753	0.752	0.673	0.829	98.080	–	–	–	–	–
realhitbench	Gemini 3.8 Flash	0.790	0.789	0.716	0.860	98.080	–	–	–	–	–
realhitbench	GPT-5.4-mini	0.715	0.709	0.621	0.789	98.080	–	–	–	–	–
realhitbench	GPT-5.5	0.768	0.764	0.691	0.836	98.080	–	–	–	–	–
realhitbench	gpt-oss-120b	0.652	0.653	0.568	0.742	98.080	–	–	–	–	–
realhitbench	Grok 4	0.686	0.684	0.601	0.768	98.080	–	–	–	–	–
realhitbench	Llama 3.3 70B	0.640	0.647	0.561	0.731	98.080	–	–	–	–	–
smat	Claude Opus 5	0.912	0.911	0.853	0.957	98.100	–	–	–	–	–

smat	Claude Sonnet 5	0.920	0.918	0.864	0.964	98.100	–	–	–	–	–
smat	DeepSeek-chat	0.895	0.892	0.827	0.946	98.100	–	–	–	–	–
smat	DeepSeek-R1	0.911	0.909	0.852	0.958	98.100	–	–	–	–	–
smat	Gemini 3.1 Pro	0.848	0.846	0.771	0.909	98.100	–	–	–	–	–
smat	Gemini 3.8 Flash	0.894	0.892	0.826	0.945	98.100	–	–	–	–	–
smat	GPT-5.4-mini	0.940	0.943	0.898	0.982	98.100	–	–	–	–	–
smat	GPT-5.5	0.921	0.919	0.868	0.965	98.100	–	–	–	–	–
smat	gpt-oss-120b	0.893	0.888	0.826	0.944	98.100	–	–	–	–	–
smat	Grok 4	0.919	0.916	0.858	0.963	98.100	–	–	–	–	–
smat	Llama 3.3 70B	0.728	0.723	0.635	0.800	98.100	–	–	–	–	–
suc	Claude Opus 5	0.998	0.998	0.994	1.000	98.920	–	–	–	–	–
suc	Claude Sonnet 5	0.950	0.948	0.914	0.979	98.920	–	–	–	–	–
suc	DeepSeek-chat	0.907	0.909	0.858	0.951	98.920	–	–	–	–	–
suc	DeepSeek-R1	0.974	0.972	0.949	0.992	98.920	–	–	–	–	–
suc	Gemini 3.1 Pro	0.969	0.965	0.936	0.988	98.920	–	–	–	–	–
suc	Gemini 3.8 Flash	0.988	0.988	0.969	1.000	98.920	–	–	–	–	–
suc	GPT-5.4-mini	0.957	0.956	0.923	0.984	98.920	–	–	–	–	–
suc	GPT-5.5	0.973	0.972	0.945	0.992	98.920	–	–	–	–	–
suc	gpt-oss-120b	0.957	0.956	0.917	0.985	98.920	–	–	–	–	–
suc	Grok 4	0.968	0.965	0.937	0.991	98.920	–	–	–	–	–
suc	Llama 3.3 70B	0.833	0.836	0.775	0.892	98.920	–	–	–	–	–
tab_fact	Claude Opus 5	0.660	0.667	0.571	0.762	99.360	0.673	0.696	0.586	0.781	0.500
tab_fact	Claude Sonnet 5	0.740	0.733	0.641	0.820	99.360	0.698	0.710	0.590	0.806	0.500
tab_fact	DeepSeek-chat	0.840	0.826	0.744	0.901	99.360	0.789	0.789	0.677	0.884	0.500
tab_fact	DeepSeek-R1	0.770	0.763	0.672	0.846	99.360	0.729	0.740	0.617	0.836	0.500
tab_fact	Gemini 3.1 Pro	0.830	0.818	0.746	0.901	99.360	0.785	0.788	0.672	0.876	0.500
tab_fact	Gemini 3.8 Flash	0.860	0.848	0.773	0.913	99.360	0.821	0.821	0.720	0.900	0.500
tab_fact	GPT-5.4-mini	0.820	0.808	0.734	0.891	99.360	0.769	0.772	0.654	0.859	0.500
tab_fact	GPT-5.5	0.840	0.829	0.748	0.903	99.360	0.800	0.803	0.700	0.892	0.500
tab_fact	gpt-oss-120b	0.830	0.815	0.739	0.890	99.360	0.773	0.773	0.647	0.868	0.500
tab_fact	Grok 4	0.850	0.839	0.762	0.912	99.360	0.810	0.812	0.707	0.897	0.500
tab_fact	Llama 3.3 70B	0.780	0.769	0.681	0.853	99.360	0.725	0.732	0.615	0.832	0.500
tabis	Claude Opus 5	0.918	0.923	0.874	0.954	77.069	–	–	–	–	–
tabis	Claude Sonnet 5	0.879	0.877	0.810	0.932	77.069	–	–	–	–	–
tabis	DeepSeek-chat	0.743	0.689	0.584	0.782	77.069	–	–	–	–	–
tabis	DeepSeek-R1	0.927	0.939	0.914	0.957	77.069	–	–	–	–	–
tabis	Gemini 3.1 Pro	0.908	0.928	0.898	0.953	77.069	–	–	–	–	–
tabis	Gemini 3.8 Flash	0.908	0.928	0.898	0.953	77.069	–	–	–	–	–
tabis	GPT-5.4-mini	0.862	0.867	0.803	0.921	77.069	–	–	–	–	–
tabis	GPT-5.5	0.918	0.934	0.905	0.956	77.069	–	–	–	–	–

tabis	gpt-oss-120b	0.880	0.911	0.874	0.943	77.069	–	–	–	–	–
tabis	Grok 4	0.938	0.946	0.924	0.961	77.069	–	–	–	–	–
tabis	Llama 3.3 70B	0.785	0.786	0.703	0.863	77.069	–	–	–	–	–
tablebench	Claude Opus 5	0.883	0.887	0.829	0.945	89.187	–	–	–	–	–
tablebench	Claude Sonnet 5	0.901	0.906	0.856	0.951	89.187	–	–	–	–	–
tablebench	DeepSeek-chat	0.869	0.884	0.828	0.934	89.187	–	–	–	–	–
tablebench	DeepSeek-R1	0.848	0.858	0.787	0.914	89.187	–	–	–	–	–
tablebench	Gemini 3.1 Pro	0.884	0.886	0.826	0.943	89.187	–	–	–	–	–
tablebench	Gemini 3.8 Flash	0.882	0.886	0.824	0.937	89.187	–	–	–	–	–
tablebench	GPT-5.4-mini	0.873	0.876	0.820	0.929	89.187	–	–	–	–	–
tablebench	GPT-5.5	0.883	0.886	0.825	0.942	89.187	–	–	–	–	–
tablebench	gpt-oss-120b	0.855	0.859	0.798	0.916	89.187	–	–	–	–	–
tablebench	Grok 4	0.853	0.859	0.789	0.918	89.187	–	–	–	–	–
tablebench	Llama 3.3 70B	0.728	0.728	0.650	0.802	89.187	–	–	–	–	–
tableeval	Claude Opus 5	0.781	0.798	0.704	0.885	69.689	–	–	–	–	–
tableeval	Claude Sonnet 5	0.759	0.789	0.693	0.876	69.689	–	–	–	–	–
tableeval	DeepSeek-chat	0.741	0.737	0.642	0.828	69.689	–	–	–	–	–
tableeval	DeepSeek-R1	0.788	0.772	0.674	0.866	69.689	–	–	–	–	–
tableeval	Gemini 3.1 Pro	0.796	0.806	0.709	0.897	69.689	–	–	–	–	–
tableeval	Gemini 3.8 Flash	0.788	0.808	0.713	0.893	69.689	–	–	–	–	–
tableeval	GPT-5.4-mini	0.775	0.784	0.685	0.871	69.689	–	–	–	–	–
tableeval	GPT-5.5	0.800	0.805	0.706	0.893	69.689	–	–	–	–	–
tableeval	gpt-oss-120b	0.795	0.793	0.699	0.883	69.689	–	–	–	–	–
tableeval	Grok 4	0.806	0.794	0.692	0.886	69.689	–	–	–	–	–
tableeval	Llama 3.3 70B	0.694	0.677	0.568	0.775	69.689	–	–	–	–	–
tat_qa	Claude Opus 5	0.833	0.801	0.708	0.886	60.725	–	–	–	–	–
tat_qa	Claude Sonnet 5	0.829	0.808	0.726	0.887	60.725	–	–	–	–	–
tat_qa	DeepSeek-chat	0.792	0.749	0.646	0.843	60.725	–	–	–	–	–
tat_qa	DeepSeek-R1	0.791	0.756	0.659	0.853	60.725	–	–	–	–	–
tat_qa	Gemini 3.1 Pro	0.795	0.750	0.654	0.833	60.725	–	–	–	–	–
tat_qa	Gemini 3.8 Flash	0.806	0.776	0.677	0.861	60.725	–	–	–	–	–
tat_qa	GPT-5.4-mini	0.795	0.748	0.661	0.828	60.725	–	–	–	–	–
tat_qa	GPT-5.5	0.803	0.768	0.684	0.855	60.725	–	–	–	–	–
tat_qa	gpt-oss-120b	0.780	0.722	0.622	0.814	60.725	–	–	–	–	–
tat_qa	Grok 4	0.792	0.733	0.637	0.821	60.725	–	–	–	–	–
tat_qa	Llama 3.3 70B	0.757	0.699	0.588	0.796	60.725	–	–	–	–	–
tpcdi_cells	Claude Opus 5	0.422	0.452	0.364	0.535	95.091	–	–	–	–	–
tpcdi_cells	Claude Sonnet 5	0.422	0.452	0.366	0.542	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-chat	0.422	0.452	0.366	0.541	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-R1	0.424	0.453	0.365	0.540	95.091	–	–	–	–	–
tpcdi_cells	Gemini 3.1 Pro	0.422	0.452	0.361	0.534	95.091	–	–	–	–	–

tpcdi_cells	Gemini 3.8 Flash	0.422	0.452	0.362	0.541	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.4-mini	0.420	0.449	0.362	0.529	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.5	0.424	0.453	0.368	0.541	95.091	–	–	–	–	–
tpcdi_cells	gpt-oss-120b	0.404	0.440	0.350	0.519	95.091	–	–	–	–	–
tpcdi_cells	Grok 4	0.404	0.440	0.358	0.522	95.091	–	–	–	–	–
tpcdi_cells	Llama 3.3 70B	0.403	0.440	0.354	0.531	95.091	–	–	–	–	–
valentine	Claude Opus 5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Claude Sonnet 5	0.991	0.985	0.952	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-chat	0.974	0.958	0.908	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-R1	0.991	0.987	0.960	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.1 Pro	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.8 Flash	0.970	0.988	0.973	1.000	81.736	–	–	–	–	–
valentine	GPT-5.4-mini	0.982	0.972	0.929	1.000	81.736	–	–	–	–	–
valentine	GPT-5.5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	gpt-oss-120b	0.973	0.961	0.915	1.000	81.736	–	–	–	–	–
valentine	Grok 4	0.966	0.944	0.884	1.000	81.736	–	–	–	–	–
valentine	Llama 3.3 70B	0.965	0.949	0.896	0.990	81.736	–	–	–	–	–
walmart_amazon	Claude Opus 5	0.980	0.982	0.947	1.000	70.807	0.976	0.912	0.770	1.000	0.094
walmart_amazon	Claude Sonnet 5	0.950	0.962	0.914	0.996	70.807	0.938	0.823	0.644	0.977	0.094
walmart_amazon	DeepSeek-chat	0.950	0.962	0.914	0.996	70.807	0.938	0.823	0.661	0.977	0.094
walmart_amazon	DeepSeek-R1	0.960	0.978	0.940	0.998	70.807	0.950	0.887	0.733	0.989	0.094
walmart_amazon	Gemini 3.1 Pro	0.970	0.967	0.920	1.000	70.807	0.964	0.848	0.669	1.000	0.094
walmart_amazon	Gemini 3.8 Flash	0.990	0.998	0.993	1.000	70.807	0.988	0.988	0.955	1.000	0.094
walmart_amazon	GPT-5.4-mini	0.950	0.975	0.941	0.998	70.807	0.937	0.874	0.708	0.978	0.094
walmart_amazon	GPT-5.5	0.970	0.967	0.920	1.000	70.807	0.964	0.848	0.676	1.000	0.094
walmart_amazon	gpt-oss-120b	0.940	0.973	0.936	0.996	70.807	0.923	0.860	0.690	0.974	0.094
walmart_amazon	Grok 4	0.960	0.978	0.943	0.998	70.807	0.950	0.887	0.733	0.989	0.094
walmart_amazon	Llama 3.3 70B	0.930	0.971	0.935	0.994	70.807	0.909	0.847	0.703	0.966	0.094
wdc_lspc	Claude Opus 5	0.990	0.996	0.986	1.000	75.082	0.991	0.992	0.974	1.000	0.264
wdc_lspc	Claude Sonnet 5	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.957	1.000	0.264
wdc_lspc	DeepSeek-chat	0.960	0.940	0.876	0.995	75.082	0.964	0.895	0.785	0.989	0.264
wdc_lspc	DeepSeek-R1	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Gemini 3.1 Pro	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.959	1.000	0.264
wdc_lspc	Gemini 3.8 Flash	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.959	1.000	0.264
wdc_lspc	GPT-5.4-mini	0.970	0.986	0.967	1.000	75.082	0.971	0.973	0.937	1.000	0.264
wdc_lspc	GPT-5.5	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	gpt-oss-120b	0.980	0.990	0.976	1.000	75.082	0.981	0.981	0.951	1.000	0.264
wdc_lspc	Grok 4	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Llama 3.3 70B	0.960	0.940	0.875	0.994	75.082	0.964	0.895	0.784	0.990	0.264
wdc_products	Claude Opus 5	0.980	0.990	0.975	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Claude Sonnet 5	0.970	0.986	0.967	1.000	77.046	0.971	0.974	0.938	1.000	0.280

wdc_products	DeepSeek-chat	0.970	0.977	0.941	1.000	77.046	0.972	0.958	0.905	1.000	0.280
wdc_products	DeepSeek-R1	0.980	0.990	0.975	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.1 Pro	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.8 Flash	0.990	0.995	0.984	1.000	77.046	0.991	0.991	0.967	1.000	0.280
wdc_products	GPT-5.4-mini	0.970	0.986	0.965	1.000	77.046	0.971	0.974	0.935	1.000	0.280
wdc_products	GPT-5.5	0.980	0.979	0.939	1.000	77.046	0.981	0.964	0.895	1.000	0.280
wdc_products	gpt-oss-120b	0.980	0.979	0.939	1.000	77.046	0.981	0.964	0.899	1.000	0.280
wdc_products	Grok 4	0.984	0.992	0.978	1.000	77.046	0.985	0.985	0.961	1.000	0.280
wdc_products	Llama 3.3 70B	0.970	0.977	0.942	1.000	77.046	0.972	0.958	0.899	1.000	0.280
wikitablequestions	Claude Opus 5	0.883	0.882	0.818	0.939	97.686	–	–	–	–	–
wikitablequestions	Claude Sonnet 5	0.867	0.866	0.803	0.927	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-chat	0.848	0.847	0.775	0.905	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-R1	0.861	0.860	0.795	0.923	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.1 Pro	0.855	0.855	0.787	0.913	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.8 Flash	0.882	0.882	0.820	0.938	97.686	–	–	–	–	–
wikitablequestions	GPT-5.4-mini	0.853	0.851	0.781	0.910	97.686	–	–	–	–	–
wikitablequestions	GPT-5.5	0.864	0.862	0.797	0.918	97.686	–	–	–	–	–
wikitablequestions	gpt-oss-120b	0.826	0.823	0.751	0.890	97.686	–	–	–	–	–
wikitablequestions	Grok 4	0.867	0.865	0.798	0.926	97.686	–	–	–	–	–
wikitablequestions	Llama 3.3 70B	0.759	0.758	0.680	0.834	97.686	–	–	–	–	–

Table 26: Official-split estimates under the headline rule averaged over the ten models.

dataset	suite_mean	official_ estimate	suite_f1_ pos	official_precision_ pos	official_recall_ pos	official_f1_ pos	official_positive_ share
abt_buy	0.971	0.987	0.969	0.933	0.950	0.940	0.108
alaska_camera	0.945	0.945	0.939	1.000	0.895	0.939	0.500
alaska_schema	0.911	0.911	–	–	–	–	–
amazon_google	0.935	0.932	0.927	0.609	0.940	0.737	0.102
bird	0.255	0.274	–	–	–	–	–
dblp_acm	0.987	0.992	0.986	0.981	0.980	0.978	0.180
dblp_scholar	0.948	0.966	0.947	0.901	0.922	0.909	0.186
fetaqa	0.555	0.555	–	–	–	–	–
fintagging	0.830	0.830	–	–	–	–	–
hitab	0.776	0.790	–	–	–	–	–
machamp	0.887	0.891	0.892	0.649	0.852	0.735	0.175
magneto_gdc	0.919	0.922	–	–	–	–	–
mmtu	0.867	0.816	–	–	–	–	–
officeqa	0.286	0.286	–	–	–	–	–
officeqa_pro_v2	0.312	0.312	–	–	–	–	–
opensanctions_ pairs	0.835	0.748	0.790	0.993	0.677	0.795	0.769
papadakis_dn	0.940	0.987	0.831	0.527	0.967	0.662	0.018
realhitbench	0.721	0.720	–	–	–	–	–
smat	0.889	0.887	–	–	–	–	–
suc	0.952	0.951	–	–	–	–	–
tab_fact	0.802	0.792	0.761	0.903	0.674	0.767	0.500
tabis	0.879	0.884	–	–	–	–	–
tablebench	0.860	0.865	–	–	–	–	–
tableeval	0.775	0.778	–	–	–	–	–
tat_qa	0.798	0.756	–	–	–	–	–
tpcdi_cells	0.417	0.449	–	–	–	–	–
valentine	0.983	0.977	–	–	–	–	–
walmart_amazon	0.959	0.974	0.949	0.824	0.931	0.872	0.094
wdc_lspc	0.982	0.984	0.983	0.968	0.978	0.972	0.264
wdc_products	0.978	0.985	0.979	0.982	0.967	0.974	0.280
wikitablequestions	0.851	0.850	–	–	–	–	–

Table 27: Cross-dataset pairwise tests under rule exact match: Friedman $\chi^2 = 92.8$ ($p = 1.5e\text{-}15$), Iman–Davenport $F = 12.8$, Nemenyi CD = 2.71, Kendall's $W = 0.281$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4677	21	7	3	0.0223	0.6458	no	no
Claude Opus 5	DeepSeek-chat	-3.0968	22	6	3	0.0047	0.1852	yes	no
Claude Opus 5	DeepSeek-R1	-1.3387	15	8	8	0.0706	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.0323	21	6	4	0.0096	0.3347	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0484	19	8	4	0.0948	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.2581	25	5	1	0.0052	0.1969	yes	no
Claude Opus 5	GPT-5.5	-0.3710	13	8	10	0.2148	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.7903	26	3	2	0.0004	0.0157	yes	yes
Claude Opus 5	Grok 4	-2.0161	18	9	4	0.0444	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.5968	26	5	0	0.0001	0.0051	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.6290	19	5	7	0.0107	0.3640	no	no
Claude Sonnet 5	DeepSeek-R1	0.1290	13	14	4	0.4917	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.5645	15	12	4	0.7536	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.4194	11	13	7	0.3613	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.7903	19	7	5	0.1117	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.0968	7	15	9	0.0478	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.3226	22	7	2	0.0079	0.2838	yes	no
Claude Sonnet 5	Grok 4	-0.5484	18	9	4	0.4100	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-4.1290	25	3	3	0.0001	0.0034	yes	yes
DeepSeek-chat	DeepSeek-R1	1.7581	8	20	3	0.0147	0.4551	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0645	10	16	5	0.1359	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.0484	6	19	6	0.0124	0.4080	no	no
DeepSeek-chat	GPT-5.4-mini	-0.1613	12	15	4	0.2981	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.7258	6	19	6	0.0040	0.1619	yes	no
DeepSeek-chat	gpt-oss-120b	-1.6935	19	9	3	0.0995	1.0000	no	no
DeepSeek-chat	Grok 4	1.0806	12	15	4	0.2352	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.5000	22	3	6	0.0001	0.0065	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.6935	15	11	5	0.6442	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.2903	12	15	4	0.3934	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.9194	19	9	3	0.1750	1.0000	no	no
DeepSeek-R1	GPT-5.5	0.9677	8	17	6	0.1032	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.4516	26	3	2	0.0001	0.0057	yes	yes
DeepSeek-R1	Grok 4	-0.6774	14	11	6	0.4090	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.2581	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	0.9839	7	17	7	0.0544	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2258	19	7	5	0.0452	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	1.6613	6	17	8	0.0283	0.7919	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.7581	21	3	7	0.0008	0.0335	yes	yes
Gemini 3.1 Pro	Grok 4	0.0161	12	13	6	0.8136	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5645	27	4	0	0.0001	0.0037	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.2097	21	8	2	0.0395	1.0000	no	no
Gemini 3.8 Flash	GPT-5.5	0.6774	10	15	6	0.7757	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.7419	24	2	5	0.0001	0.0037	yes	yes
Gemini 3.8 Flash	Grok 4	-0.9677	16	9	6	0.1354	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5484	26	4	1	0.0001	0.0031	yes	yes
GPT-5.4-mini	GPT-5.5	2.8871	5	24	2	0.0052	0.1969	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.5323	20	10	1	0.0134	0.4301	no	no
GPT-5.4-mini	Grok 4	1.2419	10	17	4	0.4266	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3387	23	4	4	0.0001	0.0051	no	yes
GPT-5.5	gpt-oss-120b	-4.4194	27	1	3	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.6452	14	6	11	0.0598	1.0000	no	no
GPT-5.5	Llama 3.3 70B	-5.2258	27	3	1	0.0000	0.0006	yes	yes
gpt-oss-120b	Grok 4	2.7742	3	24	4	0.0002	0.0069	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.8065	20	8	3	0.0167	0.5019	no	no

Grok 4	Llama 3.3 70B	-3.5806	24	4	3	0.0001	0.0034	yes	yes

Table 28: Cross-dataset pairwise tests under rule ensemble: Friedman $\chi^2 = 82.1$ ($p = 2.0\text{e-}13$), Iman–Davenport $F = 10.8$, Nemenyi CD = 2.71, Kendall's $W = 0.257$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5161	21	8	2	0.0275	0.7962	no	no
Claude Opus 5	DeepSeek-chat	-2.9839	23	7	1	0.0049	0.1720	yes	no
Claude Opus 5	DeepSeek-R1	-1.5645	17	11	3	0.0512	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.9355	22	6	3	0.0043	0.1581	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5645	18	11	2	0.2942	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.7419	25	6	0	0.0043	0.1581	yes	no
Claude Opus 5	GPT-5.5	-0.0484	14	10	7	0.2469	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5968	26	4	1	0.0003	0.0127	yes	yes
Claude Opus 5	Grok 4	-1.9839	18	10	3	0.0311	0.8394	no	no
Claude Opus 5	Llama 3.3 70B	-5.1290	25	6	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.4677	20	7	4	0.0318	0.8394	no	no
Claude Sonnet 5	DeepSeek-R1	-0.0484	14	15	2	0.9531	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.4194	15	14	2	0.6241	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9516	11	17	3	0.1215	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.2258	19	9	3	0.0712	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.4677	10	18	3	0.0611	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.0806	21	9	1	0.0070	0.2325	yes	no
Claude Sonnet 5	Grok 4	-0.4677	18	11	2	0.2767	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6129	25	4	2	0.0001	0.0030	yes	yes
DeepSeek-chat	DeepSeek-R1	1.4194	11	19	1	0.1055	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0484	11	18	2	0.2395	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.4194	7	21	3	0.0054	0.1827	no	no
DeepSeek-chat	GPT-5.4-mini	0.2419	10	17	4	0.2168	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.9355	6	21	4	0.0033	0.1276	yes	no
DeepSeek-chat	gpt-oss-120b	-1.6129	21	9	1	0.0761	1.0000	no	no
DeepSeek-chat	Grok 4	1.0000	12	16	3	0.2943	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1452	22	5	4	0.0009	0.0361	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.3710	16	14	1	0.6239	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	1.0000	11	18	2	0.1442	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.1774	19	11	1	0.2166	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.5161	8	19	4	0.0465	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.0323	25	5	1	0.0001	0.0035	yes	yes
DeepSeek-R1	Grok 4	-0.4194	13	14	4	0.7389	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.5645	24	5	2	0.0000	0.0013	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.3710	9	17	5	0.0212	0.6362	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-0.8065	20	8	3	0.1415	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	1.8871	5	22	4	0.0009	0.0368	no	yes
Gemini 3.1 Pro	gpt-oss-120b	-2.6613	22	5	4	0.0009	0.0366	no	yes
Gemini 3.1 Pro	Grok 4	-0.0484	14	15	2	0.9219	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.1935	25	6	0	0.0002	0.0083	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.1774	22	8	1	0.0288	0.8062	no	no
Gemini 3.8 Flash	GPT-5.5	0.5161	13	16	2	0.8064	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.0323	26	4	1	0.0001	0.0038	yes	yes
Gemini 3.8 Flash	Grok 4	-1.4194	21	9	1	0.0625	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5645	26	4	1	0.0001	0.0037	yes	yes
GPT-5.4-mini	GPT-5.5	2.6935	6	24	1	0.0118	0.3765	no	no
GPT-5.4-mini	gpt-oss-120b	-1.8548	23	8	0	0.0040	0.1504	no	no
GPT-5.4-mini	Grok 4	0.7581	13	15	3	0.9375	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3871	25	4	2	0.0000	0.0021	no	yes
GPT-5.5	gpt-oss-120b	-4.5484	27	2	2	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.9355	19	8	4	0.0124	0.3853	no	no
GPT-5.5	Llama 3.3 70B	-5.0806	27	4	0	0.0000	0.0011	yes	yes
gpt-oss-120b	Grok 4	2.6129	6	25	0	0.0002	0.0084	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.5323	20	11	0	0.1191	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.1452	24	6	1	0.0005	0.0233	yes	yes

Table 29: Cross-dataset pairwise tests under rule native metric: Friedman $\chi^2 = 90.2$ ($p = 4.9\text{e-}15$), Iman–Davenport $F = 12.3$, Nemenyi CD = 2.71, Kendall's $W = 0.275$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2903	20	7	4	0.0240	0.7211	no	no
Claude Opus 5	DeepSeek-chat	-2.9194	21	7	3	0.0094	0.3276	yes	no
Claude Opus 5	DeepSeek-R1	-1.2742	15	10	6	0.1094	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5484	20	7	4	0.0421	1.0000	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5161	15	11	5	0.4315	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.7581	24	5	2	0.0055	0.2039	yes	no
Claude Opus 5	GPT-5.5	0.1290	12	10	9	0.4849	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.4839	26	4	1	0.0005	0.0221	yes	yes
Claude Opus 5	Grok 4	-1.7258	18	9	4	0.0624	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.2581	26	5	0	0.0001	0.0046	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.6290	20	6	5	0.0240	0.7211	no	no
Claude Sonnet 5	DeepSeek-R1	0.0161	12	14	5	0.5231	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.2581	14	13	4	0.9219	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.7742	11	15	5	0.2429	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4677	20	7	4	0.0554	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.4194	8	17	6	0.0141	0.4644	no	no
Claude Sonnet 5	gpt-oss-120b	-3.1935	22	7	2	0.0037	0.1444	yes	no
Claude Sonnet 5	Grok 4	-0.4355	16	10	5	0.5760	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.9677	25	4	2	0.0001	0.0026	yes	yes
DeepSeek-chat	DeepSeek-R1	1.6452	9	19	3	0.0284	0.7657	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.3710	9	18	4	0.0571	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.4032	5	20	6	0.0062	0.2232	no	no
DeepSeek-chat	GPT-5.4-mini	0.1613	10	15	6	0.2760	1.0000	no	no
DeepSeek-chat	GPT-5.5	3.0484	4	21	6	0.0006	0.0230	yes	yes
DeepSeek-chat	gpt-oss-120b	-1.5645	18	8	5	0.0758	1.0000	no	no
DeepSeek-chat	Grok 4	1.1935	11	16	4	0.1436	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.3387	23	3	5	0.0001	0.0045	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.2742	12	13	6	0.7907	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.7581	12	16	3	0.2092	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.4839	18	9	4	0.1784	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.4032	7	19	5	0.0349	0.9084	no	no
DeepSeek-R1	gpt-oss-120b	-3.2097	25	4	2	0.0001	0.0044	yes	yes
DeepSeek-R1	Grok 4	-0.4516	12	13	6	0.7756	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.9839	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.0323	8	16	7	0.0679	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2097	18	8	5	0.0593	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	1.6774	5	17	9	0.0213	0.6616	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.9355	22	3	6	0.0005	0.0216	yes	yes
Gemini 3.1 Pro	Grok 4	-0.1774	12	12	7	0.6800	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.7097	25	4	2	0.0001	0.0035	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.2419	21	9	1	0.0394	0.9847	no	no
Gemini 3.8 Flash	GPT-5.5	0.6452	11	14	6	0.8520	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.9677	25	2	4	0.0000	0.0025	yes	yes
Gemini 3.8 Flash	Grok 4	-1.2097	19	10	2	0.0664	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.7419	26	4	1	0.0001	0.0030	yes	yes
GPT-5.4-mini	GPT-5.5	2.8871	5	25	1	0.0048	0.1843	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.7258	21	10	0	0.0124	0.4205	no	no
GPT-5.4-mini	Grok 4	1.0323	11	15	5	0.5043	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.5000	25	3	3	0.0000	0.0024	no	yes
GPT-5.5	gpt-oss-120b	-4.6129	26	1	4	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.8548	17	6	8	0.0256	0.7211	no	no
GPT-5.5	Llama 3.3 70B	-5.3871	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.7581	3	24	4	0.0001	0.0052	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.7742	19	8	4	0.0196	0.6269	no	no

Grok 4	Llama 3.3 70B	-3.5323	23	4	4	0.0001	0.0038	yes	yes

Table 30: Cross-dataset pairwise tests under rule top1: Friedman $\chi^2 = 80.4$ ($p = 4.2\text{e-}13$), Iman–Davenport $F = 10.5$, Nemenyi CD = 2.71, Kendall's $W = 0.252$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4839	21	8	2	0.0303	0.8495	no	no
Claude Opus 5	DeepSeek-chat	-3.0806	23	7	1	0.0035	0.1329	yes	no
Claude Opus 5	DeepSeek-R1	-1.5000	17	11	3	0.0489	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.1613	22	6	3	0.0028	0.1083	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.6290	18	11	2	0.3672	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.7097	25	6	0	0.0048	0.1663	no	no
Claude Opus 5	GPT-5.5	-0.1129	14	10	7	0.2163	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5323	26	4	1	0.0003	0.0148	yes	yes
Claude Opus 5	Grok 4	-2.0806	19	9	3	0.0242	0.7260	no	no
Claude Opus 5	Llama 3.3 70B	-5.1290	25	6	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5968	20	7	4	0.0274	0.7947	no	no
Claude Sonnet 5	DeepSeek-R1	-0.0161	14	15	2	0.9687	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.6774	16	13	2	0.5115	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8548	11	17	3	0.1389	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.2258	19	9	3	0.0776	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.3710	10	18	3	0.0611	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.0484	21	9	1	0.0089	0.2934	yes	no
Claude Sonnet 5	Grok 4	-0.5968	19	10	2	0.2357	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6452	25	4	2	0.0001	0.0032	yes	yes
DeepSeek-chat	DeepSeek-R1	1.5806	11	19	1	0.0695	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	0.9194	11	18	2	0.2681	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.4516	6	22	3	0.0042	0.1557	no	no
DeepSeek-chat	GPT-5.4-mini	0.3710	10	17	4	0.2395	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.9677	6	21	4	0.0048	0.1663	yes	no
DeepSeek-chat	gpt-oss-120b	-1.4516	20	10	1	0.1288	1.0000	no	no
DeepSeek-chat	Grok 4	1.0000	12	16	3	0.2766	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0484	22	5	4	0.0011	0.0461	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.6613	17	13	1	0.3937	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.8710	12	17	2	0.1794	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.2097	19	11	1	0.2166	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.3871	8	19	4	0.0499	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.0323	25	5	1	0.0002	0.0082	yes	yes
DeepSeek-R1	Grok 4	-0.5806	14	13	4	0.5240	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.6290	24	5	2	0.0000	0.0011	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.5323	9	17	5	0.0318	0.8584	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-0.5484	18	10	3	0.1640	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	2.0484	5	22	4	0.0025	0.1014	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.3710	21	6	4	0.0019	0.0775	no	no
Gemini 3.1 Pro	Grok 4	0.0806	13	16	2	0.9687	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-2.9677	25	6	0	0.0001	0.0069	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.0806	22	8	1	0.0342	0.8903	no	no
Gemini 3.8 Flash	GPT-5.5	0.5161	12	17	2	0.9375	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.9032	26	4	1	0.0001	0.0045	yes	yes
Gemini 3.8 Flash	Grok 4	-1.4516	21	9	1	0.0845	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5000	26	4	1	0.0001	0.0033	yes	yes
GPT-5.4-mini	GPT-5.5	2.5968	6	24	1	0.0224	0.6944	no	no
GPT-5.4-mini	gpt-oss-120b	-1.8226	23	8	0	0.0043	0.1564	no	no
GPT-5.4-mini	Grok 4	0.6290	13	15	3	0.8909	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4194	25	4	2	0.0000	0.0022	no	yes
GPT-5.5	gpt-oss-120b	-4.4194	26	3	2	0.0000	0.0011	yes	yes
GPT-5.5	Grok 4	-1.9677	19	8	4	0.0139	0.4439	no	no
GPT-5.5	Llama 3.3 70B	-5.0161	26	5	0	0.0000	0.0016	yes	yes
gpt-oss-120b	Grok 4	2.4516	6	25	0	0.0002	0.0099	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.5968	20	11	0	0.1416	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.0484	24	6	1	0.0006	0.0251	yes	yes

Table 31: Cross-dataset pairwise tests under rule top2: Friedman $\chi^2 = 79.8$ ($p = 5.5\text{e-}13$), Iman–Davenport $F = 10.4$, Nemenyi CD = 2.71, Kendall's $W = 0.250$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5000	21	8	2	0.0289	0.8084	no	no
Claude Opus 5	DeepSeek-chat	-3.0323	23	7	1	0.0040	0.1506	yes	no
Claude Opus 5	DeepSeek-R1	-1.4194	17	11	3	0.0535	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.0806	22	6	3	0.0029	0.1150	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5484	17	12	2	0.3673	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.6613	25	6	0	0.0051	0.1722	no	no
Claude Opus 5	GPT-5.5	-0.0806	14	11	6	0.2512	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.4839	26	4	1	0.0003	0.0148	yes	yes
Claude Opus 5	Grok 4	-2.0000	19	9	3	0.0218	0.6771	no	no
Claude Opus 5	Llama 3.3 70B	-5.0806	25	6	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5323	20	7	4	0.0261	0.7557	no	no
Claude Sonnet 5	DeepSeek-R1	0.0806	13	16	2	0.9531	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.5806	16	13	2	0.5115	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9516	11	17	3	0.1389	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.1613	19	9	3	0.1036	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.4194	10	18	3	0.0728	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-2.9839	21	9	1	0.0089	0.2934	yes	no
Claude Sonnet 5	Grok 4	-0.5000	18	11	2	0.2597	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.5806	25	4	2	0.0001	0.0032	yes	yes
DeepSeek-chat	DeepSeek-R1	1.6129	11	19	1	0.0665	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	0.9516	11	18	2	0.2597	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.4839	6	22	3	0.0042	0.1515	no	no
DeepSeek-chat	GPT-5.4-mini	0.3710	10	17	4	0.2318	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.9516	6	21	4	0.0045	0.1564	yes	no
DeepSeek-chat	gpt-oss-120b	-1.4516	20	10	1	0.1338	1.0000	no	no
DeepSeek-chat	Grok 4	1.0323	12	16	3	0.2766	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0484	22	5	4	0.0010	0.0430	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.6613	17	13	1	0.3937	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.8710	12	17	2	0.1858	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.2419	19	11	1	0.2896	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.3387	8	19	4	0.0510	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.0645	25	5	1	0.0002	0.0087	yes	yes
DeepSeek-R1	Grok 4	-0.5806	14	13	4	0.5240	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.6613	24	5	2	0.0000	0.0014	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.5323	9	17	5	0.0318	0.8584	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-0.5806	18	10	3	0.1957	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	2.0000	6	21	4	0.0035	0.1360	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.4032	21	6	4	0.0028	0.1147	no	no
Gemini 3.1 Pro	Grok 4	0.0806	13	16	2	0.9844	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.0000	25	6	0	0.0001	0.0069	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.1129	22	8	1	0.0359	0.9345	no	no
Gemini 3.8 Flash	GPT-5.5	0.4677	12	17	2	0.9375	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.9355	26	4	1	0.0001	0.0049	yes	yes
Gemini 3.8 Flash	Grok 4	-1.4516	21	9	1	0.0937	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5323	26	4	1	0.0001	0.0036	yes	yes
GPT-5.4-mini	GPT-5.5	2.5806	6	24	1	0.0248	0.7444	no	no
GPT-5.4-mini	gpt-oss-120b	-1.8226	23	8	0	0.0041	0.1511	no	no
GPT-5.4-mini	Grok 4	0.6613	13	15	3	0.9064	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4194	25	4	2	0.0000	0.0022	no	yes
GPT-5.5	gpt-oss-120b	-4.4032	26	3	2	0.0000	0.0017	yes	yes
GPT-5.5	Grok 4	-1.9194	19	8	4	0.0139	0.4439	no	no
GPT-5.5	Llama 3.3 70B	-5.0000	26	5	0	0.0000	0.0016	yes	yes
gpt-oss-120b	Grok 4	2.4839	6	25	0	0.0002	0.0099	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.5968	20	11	0	0.1416	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.0806	24	6	1	0.0006	0.0251	yes	yes

Table 32: Cross-dataset pairwise tests under rule top3: Friedman $\chi^2 = 80.4$ ($p = 4.2\text{e-}13$), Iman–Davenport $F = 10.5$, Nemenyi CD = 2.71, Kendall's $W = 0.252$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4839	21	8	2	0.0303	0.8495	no	no
Claude Opus 5	DeepSeek-chat	-3.0806	23	7	1	0.0035	0.1329	yes	no
Claude Opus 5	DeepSeek-R1	-1.5000	17	11	3	0.0512	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.1613	22	6	3	0.0028	0.1083	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.6290	18	11	2	0.3672	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.7097	25	6	0	0.0051	0.1722	no	no
Claude Opus 5	GPT-5.5	-0.1129	14	10	7	0.2163	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5323	26	4	1	0.0003	0.0148	yes	yes
Claude Opus 5	Grok 4	-2.0806	19	9	3	0.0242	0.7260	no	no
Claude Opus 5	Llama 3.3 70B	-5.1290	25	6	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5968	20	7	4	0.0274	0.7947	no	no
Claude Sonnet 5	DeepSeek-R1	-0.0161	14	15	2	1.0000	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.6774	16	13	2	0.5241	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8548	11	17	3	0.1389	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.2258	19	9	3	0.0810	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.3710	10	18	3	0.0611	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.0484	21	9	1	0.0089	0.2934	yes	no
Claude Sonnet 5	Grok 4	-0.5968	19	10	2	0.2357	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6452	25	4	2	0.0001	0.0032	yes	yes
DeepSeek-chat	DeepSeek-R1	1.5806	11	19	1	0.0695	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	0.9194	11	18	2	0.2681	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.4516	6	22	3	0.0042	0.1557	no	no
DeepSeek-chat	GPT-5.4-mini	0.3710	10	17	4	0.2395	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.9677	6	21	4	0.0048	0.1663	yes	no
DeepSeek-chat	gpt-oss-120b	-1.4516	20	10	1	0.1338	1.0000	no	no
DeepSeek-chat	Grok 4	1.0000	12	16	3	0.2766	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0484	22	5	4	0.0011	0.0461	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.6613	17	13	1	0.3937	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.8710	12	17	2	0.1794	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.2097	19	11	1	0.2166	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.3871	8	19	4	0.0522	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.0323	25	5	1	0.0002	0.0082	yes	yes
DeepSeek-R1	Grok 4	-0.5806	14	13	4	0.5368	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.6290	24	5	2	0.0000	0.0014	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.5323	9	17	5	0.0318	0.8584	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-0.5484	18	10	3	0.1640	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	2.0484	5	22	4	0.0025	0.1014	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.3710	21	6	4	0.0025	0.1008	no	no
Gemini 3.1 Pro	Grok 4	0.0806	13	16	2	0.9844	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-2.9677	25	6	0	0.0001	0.0069	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.0806	22	8	1	0.0342	0.8903	no	no
Gemini 3.8 Flash	GPT-5.5	0.5161	12	17	2	0.9375	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.9032	26	4	1	0.0001	0.0049	yes	yes
Gemini 3.8 Flash	Grok 4	-1.4516	21	9	1	0.0881	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5000	26	4	1	0.0001	0.0036	yes	yes
GPT-5.4-mini	GPT-5.5	2.5968	6	24	1	0.0224	0.6944	no	no
GPT-5.4-mini	gpt-oss-120b	-1.8226	23	8	0	0.0043	0.1564	no	no
GPT-5.4-mini	Grok 4	0.6290	13	15	3	0.9064	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4194	25	4	2	0.0000	0.0022	no	yes
GPT-5.5	gpt-oss-120b	-4.4194	26	3	2	0.0000	0.0016	yes	yes
GPT-5.5	Grok 4	-1.9677	19	8	4	0.0139	0.4439	no	no
GPT-5.5	Llama 3.3 70B	-5.0161	26	5	0	0.0000	0.0016	yes	yes
gpt-oss-120b	Grok 4	2.4516	6	25	0	0.0002	0.0099	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.5968	20	11	0	0.1363	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.0484	24	6	1	0.0006	0.0251	yes	yes

Table 33: Cross-dataset pairwise tests under rule top4: Friedman $\chi^2 = 80.4$ ($p = 4.2\text{e-}13$), Iman–Davenport $F = 10.5$, Nemenyi CD = 2.71, Kendall's $W = 0.252$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4839	21	8	2	0.0303	0.8495	no	no
Claude Opus 5	DeepSeek-chat	-3.0806	23	7	1	0.0035	0.1329	yes	no
Claude Opus 5	DeepSeek-R1	-1.5000	17	11	3	0.0512	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.1613	22	6	3	0.0028	0.1111	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.6290	18	11	2	0.3672	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.7097	25	6	0	0.0051	0.1722	no	no
Claude Opus 5	GPT-5.5	-0.1129	14	10	7	0.2313	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5323	26	4	1	0.0003	0.0148	yes	yes
Claude Opus 5	Grok 4	-2.0806	19	9	3	0.0242	0.7502	no	no
Claude Opus 5	Llama 3.3 70B	-5.1290	25	6	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5968	20	7	4	0.0261	0.7818	no	no
Claude Sonnet 5	DeepSeek-R1	-0.0161	14	15	2	0.9844	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.6774	16	13	2	0.4989	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8548	11	17	3	0.1553	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.2258	19	9	3	0.1036	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.3710	10	18	3	0.0760	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.0484	21	9	1	0.0089	0.2934	yes	no
Claude Sonnet 5	Grok 4	-0.5968	19	10	2	0.2280	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6452	25	4	2	0.0001	0.0032	yes	yes
DeepSeek-chat	DeepSeek-R1	1.5806	11	19	1	0.0726	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	0.9194	11	18	2	0.2681	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.4516	6	22	3	0.0042	0.1557	no	no
DeepSeek-chat	GPT-5.4-mini	0.3710	10	17	4	0.2395	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.9677	6	21	4	0.0048	0.1663	yes	no
DeepSeek-chat	gpt-oss-120b	-1.4516	20	10	1	0.1338	1.0000	no	no
DeepSeek-chat	Grok 4	1.0000	12	16	3	0.2766	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0484	22	5	4	0.0011	0.0461	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.6613	17	13	1	0.3937	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.8710	12	17	2	0.1794	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.2097	19	11	1	0.2393	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.3871	8	19	4	0.0499	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.0323	25	5	1	0.0002	0.0087	yes	yes
DeepSeek-R1	Grok 4	-0.5806	14	13	4	0.5240	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.6290	24	5	2	0.0000	0.0012	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.5323	9	17	5	0.0318	0.8584	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-0.5484	18	10	3	0.1825	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	2.0484	5	22	4	0.0025	0.1039	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.3710	21	6	4	0.0030	0.1163	no	no
Gemini 3.1 Pro	Grok 4	0.0806	13	16	2	0.9844	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-2.9677	25	6	0	0.0001	0.0069	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.0806	22	8	1	0.0342	0.8903	no	no
Gemini 3.8 Flash	GPT-5.5	0.5161	12	17	2	0.9375	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.9032	26	4	1	0.0001	0.0049	yes	yes
Gemini 3.8 Flash	Grok 4	-1.4516	21	9	1	0.0881	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5000	26	4	1	0.0001	0.0036	yes	yes
GPT-5.4-mini	GPT-5.5	2.5968	6	24	1	0.0261	0.7818	no	no
GPT-5.4-mini	gpt-oss-120b	-1.8226	23	8	0	0.0043	0.1564	no	no
GPT-5.4-mini	Grok 4	0.6290	13	15	3	0.8909	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4194	25	4	2	0.0000	0.0022	no	yes
GPT-5.5	gpt-oss-120b	-4.4194	26	3	2	0.0000	0.0016	yes	yes
GPT-5.5	Grok 4	-1.9677	19	8	4	0.0131	0.4202	no	no
GPT-5.5	Llama 3.3 70B	-5.0161	26	5	0	0.0000	0.0016	yes	yes
gpt-oss-120b	Grok 4	2.4516	6	25	0	0.0003	0.0115	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.5968	20	11	0	0.1416	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.0484	24	6	1	0.0006	0.0251	yes	yes

Table 34: Cross-dataset pairwise tests under rule top5: Friedman $\chi^2 = 80.7$ ($p = 3.7\text{e-}13$), Iman–Davenport $F = 10.6$, Nemenyi CD = 2.71, Kendall's $W = 0.253$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4839	21	8	2	0.0289	0.8066	no	no
Claude Opus 5	DeepSeek-chat	-3.0484	23	7	1	0.0035	0.1329	yes	no
Claude Opus 5	DeepSeek-R1	-1.5645	17	11	3	0.0512	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.1290	22	6	3	0.0026	0.1042	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.6613	18	11	2	0.3270	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.7097	25	6	0	0.0051	0.1772	no	no
Claude Opus 5	GPT-5.5	-0.0806	14	10	7	0.2091	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5323	26	4	1	0.0003	0.0148	yes	yes
Claude Opus 5	Grok 4	-2.0484	19	9	3	0.0218	0.6771	no	no
Claude Opus 5	Llama 3.3 70B	-5.1613	25	6	0	0.0002	0.0084	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5645	20	7	4	0.0288	0.8066	no	no
Claude Sonnet 5	DeepSeek-R1	-0.0806	14	15	2	0.9844	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.6452	16	13	2	0.4989	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8226	11	17	3	0.1389	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.2258	19	9	3	0.0810	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.4032	10	18	3	0.0667	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.0484	21	9	1	0.0089	0.2934	yes	no
Claude Sonnet 5	Grok 4	-0.5645	19	10	2	0.2357	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6774	25	4	2	0.0001	0.0030	yes	yes
DeepSeek-chat	DeepSeek-R1	1.4839	11	19	1	0.0973	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	0.9194	11	18	2	0.2681	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.3871	7	21	3	0.0054	0.1827	no	no
DeepSeek-chat	GPT-5.4-mini	0.3387	10	17	4	0.2096	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.9677	6	21	4	0.0048	0.1711	yes	no
DeepSeek-chat	gpt-oss-120b	-1.4839	20	10	1	0.1338	1.0000	no	no
DeepSeek-chat	Grok 4	1.0000	12	16	3	0.3417	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1129	22	5	4	0.0011	0.0450	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5645	16	14	1	0.7022	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.9032	12	17	2	0.1670	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.1452	19	11	1	0.3365	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.4839	8	19	4	0.0455	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-2.9677	25	5	1	0.0002	0.0102	yes	yes
DeepSeek-R1	Grok 4	-0.4839	13	14	4	0.8216	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.5968	24	5	2	0.0000	0.0022	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.4677	9	17	5	0.0274	0.7945	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-0.5806	18	10	3	0.2723	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	2.0484	5	22	4	0.0009	0.0361	no	yes
Gemini 3.1 Pro	gpt-oss-120b	-2.4032	21	6	4	0.0028	0.1091	no	no
Gemini 3.1 Pro	Grok 4	0.0806	13	16	2	0.9844	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.0323	25	6	0	0.0002	0.0100	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.0484	22	8	1	0.0359	0.9345	no	no
Gemini 3.8 Flash	GPT-5.5	0.5806	12	17	2	0.9219	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.8710	26	4	1	0.0001	0.0049	yes	yes
Gemini 3.8 Flash	Grok 4	-1.3871	21	9	1	0.0777	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5000	26	4	1	0.0001	0.0036	yes	yes
GPT-5.4-mini	GPT-5.5	2.6290	6	24	1	0.0224	0.6771	no	no
GPT-5.4-mini	gpt-oss-120b	-1.8226	23	8	0	0.0043	0.1608	no	no
GPT-5.4-mini	Grok 4	0.6613	13	15	3	0.8909	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4516	25	4	2	0.0000	0.0020	no	yes
GPT-5.5	gpt-oss-120b	-4.4516	26	3	2	0.0000	0.0016	yes	yes
GPT-5.5	Grok 4	-1.9677	19	8	4	0.0124	0.3977	no	no
GPT-5.5	Llama 3.3 70B	-5.0806	27	4	0	0.0000	0.0015	yes	yes
gpt-oss-120b	Grok 4	2.4839	6	25	0	0.0003	0.0115	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.6290	20	11	0	0.1363	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.1129	24	6	1	0.0004	0.0187	yes	yes

Table 35: Mean score per dataset and model under the headline rule (ensemble).

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	DeepSeek-R1	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.990	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.902	0.933	0.912	0.920	0.943	0.912	0.912	0.951	0.935	0.915	0.783
amazon_google	0.930	0.940	0.940	0.960	0.944	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.404	0.389	0.213	0.329	0.290	0.213	0.151	0.231	0.271	0.121	0.196
dblp_acm	1.000	1.000	1.000	1.000	0.990	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.656	0.626	0.617	0.556	0.592	0.607	0.643	0.340	0.618	0.610	0.245
fintagging	0.849	0.865	0.847	0.849	0.822	0.831	0.839	0.841	0.789	0.805	0.789
hitab	0.757	0.785	0.759	0.776	0.789	0.777	0.744	0.792	0.816	0.751	0.795
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.954	0.964	0.962	0.945	0.973	0.939	0.926	0.896	0.937	0.953	0.658
mmtu	0.891	0.883	0.896	0.883	0.835	0.896	0.885	0.879	0.857	0.822	0.810
officeqa	0.435	0.408	0.358	0.339	0.281	0.269	0.273	0.254	0.124	0.176	0.225
officeqa_pro_v2	0.413	0.502	0.311	0.322	0.256	0.267	0.263	0.322	0.280	0.218	0.277
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.940	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.790	0.777	0.768	0.715	0.729	0.753	0.686	0.715	0.705	0.652	0.640
smat	0.894	0.912	0.921	0.920	0.911	0.848	0.919	0.940	0.895	0.893	0.728
suc	0.988	0.998	0.973	0.950	0.974	0.969	0.968	0.957	0.907	0.957	0.833
tab_fact	0.860	0.660	0.840	0.740	0.770	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.908	0.918	0.918	0.879	0.927	0.908	0.938	0.862	0.743	0.880	0.785
tablebench	0.882	0.883	0.883	0.901	0.848	0.884	0.853	0.873	0.869	0.855	0.728
tableeval	0.788	0.781	0.800	0.759	0.788	0.796	0.806	0.775	0.741	0.795	0.694
tat_qa	0.806	0.833	0.803	0.829	0.791	0.795	0.792	0.795	0.792	0.780	0.757
tpcdi_cells	0.422	0.422	0.424	0.422	0.424	0.422	0.404	0.420	0.422	0.404	0.403
valentine	0.970	1.000	1.000	0.991	0.991	1.000	0.966	0.982	0.974	0.973	0.965
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.980	0.980	0.984	0.970	0.970	0.980	0.970
wikitablequestions	0.882	0.883	0.864	0.867	0.861	0.855	0.867	0.853	0.848	0.826	0.759

Table 36: Probes: mean score per anchor operator.

op	utility	n	ensemble	em	native	tok_f1	abs_err_ensemble
identity	1.000	3083	0.998	1.000	1.000	1.000	0.002
verbose	1.000	3083	0.998	1.000	1.000	1.000	0.002
unit_variant	1.000	632	0.962	0.837	1.000	0.843	0.038
tagged	1.000	3083	0.998	1.000	1.000	1.000	0.002
paraphrase	0.950	3083	0.979	0.931	0.988	0.965	0.029
near_miss_small	0.750	632	0.769	0.097	0.101	0.112	0.019
superset	0.600	1213	0.379	0.045	0.103	0.650	0.221
typo	0.600	1104	0.287	0.101	0.101	0.459	0.313
partial	0.500	1213	0.286	0.060	0.119	0.481	0.214
near_miss_medium	0.450	632	0.625	0.008	0.008	0.021	0.175
shuffled	0.400	109	0.744	0.009	0.403	0.997	0.344
hedge	0.300	2974	0.185	0.192	0.405	0.504	0.115
list_dump	0.250	1845	0.146	0.013	0.136	0.394	0.104
digit_typo	0.200	632	0.433	0.000	0.241	0.013	0.233
near_miss_large	0.150	632	0.316	0.000	0.000	0.013	0.166
sign_flip	0.050	632	0.020	0.003	0.003	0.017	0.030
magnitude	0.050	632	0.200	0.013	0.128	0.027	0.150
flip	0.000	1238	0.000	0.000	0.000	0.000	0.000
no_answer	0.000	3083	0.000	0.000	0.000	0.001	0.000
wrong_random	0.000	1845	0.107	0.041	0.046	0.105	0.107

Table 37: Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	DeepSeek-R1	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
exact match	2 (3)	1 (1)	3 (2)	7 (5)	4 (4)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	1.5e-15	0.28
native metric	1 (3)	3 (2)	2 (1)	7 (5)	5 (4)	4 (6)	6 (7)	9 (8)	8 (9)	10 (10)	11 (11)	4.9e-15	0.27
ensemble	1 (3)	2 (1)	3 (2)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	2.0e-13	0.26
top1	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	4.2e-13	0.25
top2	1 (3)	2 (1)	3 (2)	5 (5)	4 (4)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	5.5e-13	0.25
top3	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	4.2e-13	0.25
top4	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	4.2e-13	0.25
top5	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	6 (7)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	3.7e-13	0.25

Table 38: Kendall τ between the Bradley–Terry rankings of the rules.

rule	exact match	native metric	ensemble	top1	top2	top3	top4	top5
exact match	1.000	0.855	0.855	0.891	0.891	0.891	0.891	0.891
native metric	0.855	1.000	0.782	0.818	0.818	0.818	0.818	0.818
ensemble	0.855	0.782	1.000	0.964	0.964	0.964	0.964	0.964
top1	0.891	0.818	0.964	1.000	1.000	1.000	1.000	1.000
top2	0.891	0.818	0.964	1.000	1.000	1.000	1.000	1.000
top3	0.891	0.818	0.964	1.000	1.000	1.000	1.000	1.000
top4	0.891	0.818	0.964	1.000	1.000	1.000	1.000	1.000
top5	0.891	0.818	0.964	1.000	1.000	1.000	1.000	1.000

Table 39: Metric selection summary.

quantity	value
anchor	synthetic anchors (31380 answers, 20 operators)
n_metrics	100
n_constant	10
n_representatives	45
n_candidates	28
selected	["tok_f1", "num_decay_2p5", "prefix_ratio"]
selected_idx	[33, 24, 51]
nnls_weights	"tok_f1": 0.5610768920589378, "num_decay_2p5": 0.325330660908859, "prefix_ratio": 0.1135924470322032
lodo_rho_selected	0.875
lodo_rho_native_only	0.814
lodo_rho_em_only	0.831

Table 40: Forward-selection trace.

step	added	lodo_rho	gain	stopped
1	tok_f1	0.846	1.846	–
2	num_decay_2p5	0.869	0.023	–
3	prefix_ratio	0.875	0.006	–
4	em_raw	0.877	0.002	yes

Table 41: Validation summary.

quantity	value
lodo_mean_rho_lodo	0.8940
lodo_mean_rho_full	0.8949
lodo_same_selection_share	0.9032
severity_l1_shift_mean	0.0419
severity_l1_shift_max	0.0432
severity_selection_changed_share	0.0000
probe_mae_ensemble	0.1132
probe_mae_em	0.1991
probe_mae_native	0.1545
probe_mae_tok_f1	0.1440